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Behavioural Analysis of IoT Network Traffic for Device Identification and Anomaly Detection

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Abstract

The Internet of Things continues to expand at an unprecedented rate, connecting billions of devices across smart homes, healthcare systems, and industrial production environments. This growth creates a significant attack surface, and many IoT devices lack the processing capacity and built-in security mechanisms needed to defend themselves against exploitation. Current detection approaches struggle to scale effectively across large IoT environments and can miss behavioural indicators of compromise that signature-based methods are not designed to catch.

This study investigates whether a hybrid behavioural analysis framework, combining packet-level device identification with flow-level anomaly detection, could provide an effective approach to IoT network security. The CIC IoT-DIAD 2024 dataset, developed by a team at the Canadian Institute for Cybersecurity, was used as the primary testbed, containing traffic from 105 real IoT devices and 33 distinct attack types. Implementing a Random Forest classification pipeline, three progressive device identification models and two anomaly detection models were developed to simulate the workflow in an attack and detection scenario.

Device identification achieved over 99% accuracy at the category level and 95% accuracy across all 50 devices simultaneously, with performance degrading gradually and predictably as classification complexity increased. Anomaly detection was evaluated across 3 camera devices - the Arlo Q Indoor Camera, the Nest Indoor Camera, and the Netatmo Camera, using a reduced set of 9 flow-level features derived from stream, channel and jitter statistics. Six attack types were evaluated, selected to provide coverage across five attack categories: Mirai, DoS, DDoS, Reconnaissance, and Spoofing. Binary detection exceeded 98% accuracy, and multi-class attack classification across seven classes reached 96.26% accuracy for the highest-performing device, with performance scaling clearly with benign baseline traffic volume.

The findings indicate that the hybrid behavioural approach reinforces and extends the conclusions of prior work in this area. A lightweight feature representation was shown to be sufficient for strong detection performance across multiple devices and attack types, supporting the case for further research with potential for practical deployment in resource-limited IoT environments.

Keywords: IoT security, IoT device identification, IoT anomaly detection, IoT behavioural analysis, network traffic analysis

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1

Introduction

1.1 Introduction

The Internet of Things (IoT) is an evolving landscape that continues to transform smart homes, many different industries, and entire cities into an interconnected digital web. At any given moment, countless IoT devices are continuously generating and transmitting data across networks worldwide. While these devices provide convenience and efficiency, the mere size and scale of implementation create a massive potential attack surface (Rabbani et al., 2025). Vulnerable devices can be exploited using numerous cyber attack methods, potentially leading to economic or physical consequences (Gelenbe et al., 2020).

One of the main challenges in IoT security is distinguishing between normal (benign) and potentially malicious (anomalous) network traffic. IoT devices are diverse, built by different manufacturers and using various protocols. This makes it challenging to define what benign behaviour looks like across multiple devices on a network. Many of these devices also have limited resources and weak built-in security, which leaves them exposed to attacks such as Distributed Denial of Service (DDoS), spoofing, or malware infections (Neto et al., 2023). Addressing these issues is essential to improving intrusion detection and strengthening IoT security practices.

This study aims to explore the device identification and anomaly detection in IoT traffic, with a focus on the CIC IoT-DIAD 2024 dataset. This dataset contains detailed traffic analysis from over 100 real IoT devices operating under both benign and anomalous conditions. By

analyzing these traffic patterns, the research will investigate methods of identifying IoT devices on a network, how normal device behaviour can be identified, how anomalies present themselves under different types of attacks, and how this data can support device or network security strategies in real-world IoT environments.

The next section provides background and related work, introducing the fundamentals of IoT, common attacks used against IoT devices, and existing anomaly detection methods.

1.2 Background and Related Work

1.2.1 IoT Overview

One of the earliest examples often cited as an IoT device was a Coca-Cola vending machine at the Carnegie Mellon University Computer Science department in the early 1980s. David Nichols and a small team of fellow students installed a system that monitored the stock level of cans and whether newly loaded drinks had been cooled long enough to be refreshing. The project was simple, but it illustrated the basic concept of connecting a physical object to a network so that useful information could be shared without human intervention (Farooq et al., 2015; Nichols, 1993).

Modern IoT devices follow the same principle. They are physical objects, designed to collect data from their environment and communicate it across a network. These devices can either exchange information locally with other connected devices or send it out through gateways to larger data processing systems. This architecture allows IoT devices to provide real-time monitoring or automation across a wide range of domains.

In smart homes, IoT applications are seemingly endless, and can include live-feed cameras, door sensors, and smart appliances. In healthcare, connected devices can monitor vital health data in patients and support remote care. Industrial systems integrate sensors and actuators across production facilities to streamline automation, improve efficiency and ensure safety. Simply put, the use cases are diverse, and the ever-growing IoT network makes scalability a foundational characteristic.

As IoT ecosystems continue to scale from smart homes to entire cities, the number and variety of device types present a challenge. Each device may come from a different manufacturer, and different brands and types of devices can communicate using different protocols. This heterogeneity makes it essential to accurately identify and classify devices connected to the network. Device identification, also known as profiling or fingerprinting, can then assist in device management and also in detecting anomalous or malicious behaviour (Rabbani et al., 2025). For this reason, the first step in this research is to examine methods of IoT device identification.

1.2.2 Device Identification

This is a critical initial step in monitoring and detecting anomalies in an IoT ecosystem. Research shows that features like packet size, message timing and protocol usage can reliably differentiate devices, even when manufacturers use different firmware (Rabbani et al., 2025; Safi et al., 2022). Fingerprinting can help in recognizing devices and in turn, track anomalies on those devices, allowing the isolation of potentially compromised hardware before it can cause widespread harm. A single vulnerable device, left unchecked and underdefended, can lead to disastrous consequences for the whole network.

As an example, in 2016, the Mirai BotNet exploited a single vulnerable device in a network and subsequently infected over 60,000 IoT devices. The day after Mirai was released, 15,000 cyberattacks were launched, disrupting telecommunication, game servers, and several high-profile websites (Amara Korba et al., 2025; Kolias et al., 2017; Pathak et al., 2021).

Once devices have been identified, the next step in understanding their behaviour is investigating how they communicate. IoT devices rely on a wide range of communication protocols. Exploring these protocols provides context for why certain devices behave the way they do on a network, and how anomalies in their traffic patterns can reveal potential threats.

1.2.3 IoT Security Challenges and Attack Types

Security challenges largely come from limited device resources, a lack of standardization, and weak authentication mechanisms. These examples can make IoT devices easy targets for attackers (Amara Korba et al., 2025; Rabbani et al., 2025). Common threats include large-scale DDoS attacks (botnets like Mirai), reconnaissance scanning (port and OS scanning), and web-based vulnerabilities (SQLi or command injection) (Neto et al., 2023).

Given the variety of attack methods and the fact that these intrusions typically leave anomalous patterns in network traffic, it is clear that anomaly detection techniques are well-suited to uncovering emerging threats.

1.2.4 Anomaly Detection in IoT Traffic

Anomaly detection focuses on recognizing unusual or unexpected behaviour (Song et al., 2020). Recent studies show the potential of machine learning and deep learning approaches to detect abnormal events within IoT networks (Pathak et al., 2021). By identifying these anomalies from normal device behaviour, it can provide an adaptive layer of defence that can help mitigate threats in real-time and enhance the overall security posture of IoT environments.

1.2.5 Summary of Related Work

Building on these detection foundations, research in IoT security has developed along several intersecting lines. Signature-based intrusion detection systems such as Snort (Roesch, 1999) provide reliable detection of known threats but struggle with the heterogeneity and rate of change in modern IoT environments, where new attack variants can appear faster than rule sets can be updated (Kwon et al., 2022). This has pushed the field toward behavioural approaches that analyze how devices communicate instead of what known attack signatures they match.

Within behavioural detection, two strands of work are particularly relevant to this study. Packet-level analysis has shown that selected features can fingerprint IoT devices reliably, even across manufacturers and firmware versions (Safi et al., 2022). The second strand leads to flow-level analysis, which has proven effective for detecting anomalies in traffic patterns over time, with stream, channel and jitter metrics revealing deviations that static packet inspection can miss (Rabbani et al., 2025). Progress in both areas has been supported by a number of purpose-built datasets.

BoT-IoT (Koroniotis et al., 2019) captures large-scale attack traffic, including DDoS and DoS; IoT-23 (Garcia et al., 2020) provides labelled benign and malicious traffic supporting botnet and anomaly detection studies; and more recently, CIC IoT-DIAD 2024 (Rabbani et al., 2025) captures traffic from 105 real IoT devices across 33 attack types, purpose-built for hybrid detection research. Rabbani et al. also demonstrated that combining packet and flow-level analysis into a single framework outperforms either approach used alone.

Despite this progress, two gaps remain relevant to this study: scaling hybrid detection across large and diverse device populations, and doing so with lightweight feature representations that could translate to resource-constrained deployment. These gaps frame the scope of this study.

1.3 Problem Statement

The Internet of Things has grown into a convenient global ecosystem of billions of interconnected devices, and it has also introduced significant security risks. Many IoT devices are built with limited resources and minimal protection, making them easy targets for exploitation (Safi et al., 2022).

In diverse IoT environments, a wide variety of devices with different communication patterns makes it challenging to monitor network behaviour and accurately distinguish benign traffic from anomalous activity. Research that combines device identification with anomaly detection has made progress, but current approaches remain hard to scale across large device populations and often struggle with the resource-limited hardware that IoT devices typically run.

Attacks in heterogeneous IoT environments remain difficult to detect using approaches that are both accurate at scale and practical to deploy on resource-constrained hardware.

1.4 Research Objectives

The primary objective of this project is to develop and evaluate a conceptual framework for scalable IoT device identification and anomaly detection based on behavioural traffic modelling.

In order to achieve this primary objective, the following secondary objectives have been identified:

1. Investigate the security challenges and vulnerabilities represented in the CIC IoT-DIAD 2024 (Canadian Institute for Cybersecurity (CIC) (2024)) dataset.
2. Examine existing approaches to IoT device profiling and anomaly detection as applied to real network traffic
3. Analyze the characteristics of benign and malicious traffic within the dataset to distinguish device behaviours across multiple device types and attack categories.
4. Evaluate the effectiveness of a scalable detection approach using the CIC IoT-DIAD 2024 dataset as a benchmark

1.5 Scope and Limits

This project focuses on IoT device identification and anomaly detection using network traffic data from the CIC IoT-DIAD 2024 dataset (Canadian Institute for Cybersecurity (CIC) (2024)), which captures traffic from real IoT devices under both benign and attack conditions. The analysis is conducted using machine learning (ML) classifiers applied to behavioural traffic features extracted from the dataset.

Device identification is evaluated across all 50 named devices in the dataset. Unnamed entries identified only by MAC address were excluded, as behavioural fingerprinting is only meaningful when tied to a known device. Anomaly detection is evaluated as a focused case study on three camera devices (the Arlo Q Indoor Camera, the Nest Indoor Camera, and the Netatmo Camera) against six attack types selected to provide coverage across five of the dataset's attack categories. The study analyzes this data to identify patterns that fingerprint devices and recognize anomalies that may indicate malicious activity.

This research does not involve the design or deployment of an intrusion detection system. It is instead an exploration and evaluation of existing methods in device identification and anomaly detection, applied to a publicly available dataset.

1.6 Document Structure

The remainder of the report is organized as follows:

Chapter 2 Literature Review: Examines IoT security challenges, existing detection approaches, and the theoretical foundation for the hybrid behavioural framework evaluated in this study.

Chapter 3 Research Methodology: Details the dataset, classification pipeline, feature selection approach, and evaluation metrics used throughout the report.

Chapter 4 Results / Discussion: Presents the results of the device identification and anomaly detection experiments, with analysis and discussion of the findings at each stage.

Chapter 5 Conclusion: Concludes the study, summarising the research contributions, reflecting on the process, and identifying directions for future work.

2

Literature Review

The size and diversity of modern IoT network environments create a security issue that traditional detection approaches were not designed to handle. This chapter explores the existing body of research that serves as a foundation to this study, beginning with an overview of IoT networks and the security challenges that are associated with their heterogeneous nature. Then, it looks at how signature-based detection systems fall short in this context, and explores behaviour-based alternatives: packet-level and flow-level analysis and a hybrid approach that combines them.

2.1 IoT Networks and Security Background

The Internet of Things is an evolving landscape that continues to transform smart homes, many different industries, and entire cities into an interconnected digital web. At any given moment, countless IoT devices are continuously generating and transmitting data across networks worldwide.

One of the earliest examples often cited as an IoT device was a Coca-Cola vending machine at the Carnegie Mellon University Computer Science department in the early 1980s. David Nichols and a small team of fellow students installed a system that monitored the stock level of cans and whether newly loaded drinks had been cooled long enough to be refreshing. The project was simple, but it illustrated the basic concept of connecting a physical object to a network so that useful information could be shared without human intervention (Farooq et al., 2015; Nichols, 1993).

Modern IoT devices follow the same principle. They are physical objects, designed to collect data from their environment and communicate it across a network. These devices can either exchange information locally with other connected devices or send it out through gateways to larger data processing systems. This architecture allows IoT devices to provide real-time monitoring or automation across a wide range of domains.

In smart homes, IoT applications include live-feed cameras, door sensors, and robotic vacuum cleaners. In healthcare, connected devices can monitor vital health data in patients and support remote care. Industrial systems integrate sensors and actuators across production facilities to streamline automation, improve efficiency and ensure safety. Simply put, the use cases are diverse, and scalability is a foundational characteristic.

While these devices provide convenience and efficiency, the scale of implementation creates a massive potential attack surface (Rabbani et al., 2025). Vulnerable devices can be exploited using numerous cyber attack methods, potentially leading to economic or physical consequences (Gelenbe et al., 2020). These risks directly threaten the core security principles of the CIA triad: confidentiality, integrity, and availability (Al Reshan, 2021). Attackers may intercept or expose sensitive data, alter device states or transmitted information, or disrupt essential IoT services altogether. As a result, maintaining the triad is foundational in securing modern IoT ecosystems.

Securing IoT systems requires the ability to reliably understand and interpret device behaviour at scale. IoT networks are typically *heterogeneous*, meaning they consist of a variety of diverse device types. The heterogeneity can make it difficult to create fixed profiles to identify each device (Safi et al., 2022). This cause and effect leads to two of the main challenges in IoT security: Identifying devices on a network, and distinguishing between normal (benign) and potentially malicious (anomalous) network traffic (Amara Korba et al., 2025; Chatterjee & Ahmed, 2022). Many of these devices also have limited resources, with little processing power, memory and energy availability. This often results in weak built-in security, with limited capacity to support traditional host-based security mechanisms, leaving them potentially exposed to attacks (Neto et al., 2023).

Addressing these issues is essential to improving intrusion detection and strengthening the overall security posture in modern IoT security environments. A device with weak built-in security and lacking in oversight on the network is a vulnerability itself, and to the interconnected network as a whole. Understanding how to detect and respond to anomalous behaviour across a set of devices is then a foundational concern. The question is how existing detection approaches are tackling these issues, and where they fall short, which is where this review begins.

2.2 Signature-Based Intrusion Detection

Traditional *Network Intrusion Detection Systems* (NIDSs) such as **Snort**[1] represent some of the earliest and most capable signature-based defences developed for network security. Snort operates by matching packet data against a predefined library of signatures that describe known attack behaviours, allowing for the identification of previously documented threats (Roesch, 1999). While Snort is a powerful and flexible system, it can be complex to configure and maintain, and requires manual tuning and continual updates to its rule sets to remain effective (Koziol, 2003). Additionally, signature-based IDS systems' reliance on known threat signatures makes them poorly suited for detecting zero-day exploits, advanced persistent threats (APTs), and novel malware variants (Kwon et al., 2022).

Although tools like Snort are still widely used, their reliance on manual rule management and known attack patterns limits their ability to cope with the heterogeneous nature of IoT environments. This limitation is exposed, for example, when addressing the dynamic nature of botnet attacks, where threats continuously evolve and adapt to avoid detection (Amara Korba et al., 2025). As a result, signature-based NIDSs are most effective when used as part of a layered defence strategy, as they provide valuable perimeter visibility and detection of known external threats. When paired with complementary *behavioural* approaches, such as packet- and flow-based analysis, the combination can strengthen the overall security posture by addressing known attacks at the network edge and behavioural anomalies that may occur once a device has already been compromised.

2.3 Behaviour-Based Intrusion Detection

As IoT networks grow in complexity, signature-based defences have become less effective at managing the diversity of modern networks. Comparatively, behaviour-based approaches analyze the communication patterns of devices to learn how they normally operate, allowing systems to detect both known attacks more efficiently and to flag unusual deviations from expected behaviour, potentially interrupting novel or unknown attack types (Chatterjee & Ahmed, 2022; Safi et al., 2022).

Among the range of behaviour-based techniques, two approaches most relevant to this study are **packet-based** and **flow-based** analysis. These methods examine the IoT network traffic at different levels of granularity. Packet-level analysis focuses on individual transmissions and their device-specific features, and flow-level analysis records communication patterns over time to show higher-level trends. Each operates at a different level of granularity, and together they form the basis of the hybrid approach examined later in this chapter.

Recent work at the Canadian Institute for Cybersecurity (CIC) established a strong case for *combining* packet and flow-based analysis in IoT security. Rabbani et al., 2025 demonstrated

[1] https://www.snort.org(Snort Project, n.d.)

that integrating features from both traffic layers improves the ability to recognize device behaviour and detect anomalies in real time. Their study showed that hybrid feature extraction, using both low-level packet information and high-level flow statistics, outperformed single-view approaches in accuracy and false-positive reduction (Canadian Institute for Cybersecurity (CIC), 2023; Rabbani et al., 2025).

Building on this foundation, they developed a hybrid framework and associated dataset (see Chapter 3.3) that performs two related tasks: IoT device identification and anomaly detection (DIAD).

2.3.1 Packet-Based Approach of Device Identification

The packet-based part of the framework focuses on analyzing individual network packets to extract detailed behavioural information from IoT devices. Each packet contains metadata, or *features*, with some of the most common listed below in Table 2.1.

Table 2.1: Examples of Network Packet Features

Feature	Description
Header Fields	Core identifiers (SRC/DST IP, MAC, Ports) showing where the packet comes from and where it is going.
Sequence Numbers	Track packet order and help detect missing packets or delays.
Timestamps	Record when each packet is sent or received; useful for analysing timing and frequency patterns.
Protocol Information	Indicates which network protocol is used and how the device communicates.
Packet Size	Total bytes of each packet, which may vary by device type.
HTTP Handshake Features	Elements exchanged during session startup (e.g., SSL/TLS) that reveal device behaviour.
User-Agent Strings	Text identifiers describing software, version, OS, or device model.

The combination of this data can form a unique *behavioural fingerprint* of each device's communication pattern. Some features, such as HTTPS handshakes and user-agent strings in particular, can be especially descriptive and can be used to differentiate between specific device types (Rabbani et al., 2025).

By examining these packet-level features, researchers can build a detailed profile of *what type* of device is communicating. This information supports both accurate device identification and the recording of a baseline benign state, against which future activity can be compared. Divergences from these expected patterns may indicate potential compromise, malfunctions, or misconfigurations on the devices themselves. This detailed view makes it

effective for identifying device fingerprints and recognizing early indicators of compromise (IOC) in encrypted (Song et al., 2020) or environments that use multiple protocols (Safi et al., 2022).

2.3.2 Flow-Based Approach for Anomaly Detection

In the flow-based approach, analysis focuses on sequences of packets exchanged between hosts over time, rather than inspecting individual packets as in the packet-based approach. This method captures broader patterns of communication that describe *how* devices interact within the network. These patterns can also be compared against a recorded baseline benign state to identify anomalies. A significant deviation from the established normal flow behaviour may indicate network instability or early stages of an attack.

Flow-based analysis has been widely applied in past research. For example, in 2018, M. Alkasassbeh proposed a hybrid prediction-based model that uses SNMP-MIB variables to predict expected traffic that could identify volume-based deviations that point-in-time packet inspection might miss (Alkasassbeh, 2018). The temporal approach has been further refined through the monitoring of specialized calculation features (including stream counts, communication channels and packet arrival jitter) to identify trend changes characteristic of threats like DDoS and botnet activity (Rabbani et al., 2025). Studies such as these demonstrate the effectiveness of flow-level features in revealing anomalies that may not be visible at the packet level.

To quantify these patterns at the flow level, we can expand on the mentioned three primary metrics, named *Stream*, *Channel* and *Jitter* (Table 2.2), which are commonly used in IoT traffic analysis.

Table 2.2: Flow-based behavioural metrics used for anomaly detection (Rabbani et al., 2025).

Metric	Description
Stream Metrics	Describe how devices send data within a flow, including burst duration and gaps between packets. These patterns characterize normal communication rhythm.
Channel Metrics	Capture the direction and balance of traffic between devices, indicating whether communication is one-way or bidirectional. Sudden imbalances may reflect scanning or flooding activity.
Jitter Metrics	Measure short-term variations in packet delay within a stream, typically over windows of 5, 10, 30, or 60 seconds. Elevated jitter can reveal congestion or early-stage attacks.

Together, these flow-based features can help visualize timing and stability in device communications, making it possible for a model to detect unusual activity without needing to inspect every packet's contents. Packet-level analysis captures device detail but misses broader behavioural trends, while flow-level analysis reveals patterns over time but lacks the granularity

to fingerprint individual devices. Combining both layers is what gives the hybrid framework its range, and it is that combination which the following section examines.

For reference, both packet- and flow-analysis methods are summarised in Table 2.3.

Table 2.3: Packet-based vs flow-based approaches to data analysis

Feature Extraction Method	Core Focus	Key Feature Examples	Purpose in Framework
Packet-based approach	Capturing individual packet characteristics and security state	HTTPS-related features, handshake information, user-agent strings, packet length, hash values	Primarily for IoT device identification and analysis of encrypted communications
Flow-based approach	Analyzing sequences of packets (flows) over time	*Stream*, *channel* and *jitter* metrics calculated over various time intervals (e.g. 1, 5, 10, 30 and 60 seconds)	Primarily for anomaly detection, distinguishing malicious activity based on timing and volume patterns

2.4 Packet + Flow = Hybrid Approach

The integration of packet and flow-based analysis forms the foundation of a hybrid IoT intrusion-detection security framework. The Device Identification and Anomaly Detection (DIAD) hybrid framework proposed by Rabbani et al. (2025), shown in Figure 2.1, visualizes this approach:

The system begins by capturing network traffic from IoT devices and extracting relevant features from packet capture (pcap) files. These features are then converted into CSV format, normalized, and reduced to select the most informative attributes for analysis.

In this framework, the device identification phase uses a machine learning (ML) classifier to label devices based on behavioural features, while the anomaly detection phase evaluates these behaviours to determine whether the activity is benign or malicious. The secondary classification component further categorizes the *type* of detected attack. The dataset covers seven broad attack categories (Neto et al., 2023), each representing a distinct method of disrupting or exploiting IoT devices:

- **DDoS** — large volumes of traffic from multiple compromised devices flood a target simultaneously, overwhelming its ability to function normally.
- **DoS** — the same principle as DDoS, but carried out from a single source, typically targeting one device or service directly.
- **Reconnaissance** — scanning and probing activity used to map a network, identify open ports, and locate potential weaknesses before a more destructive attack.

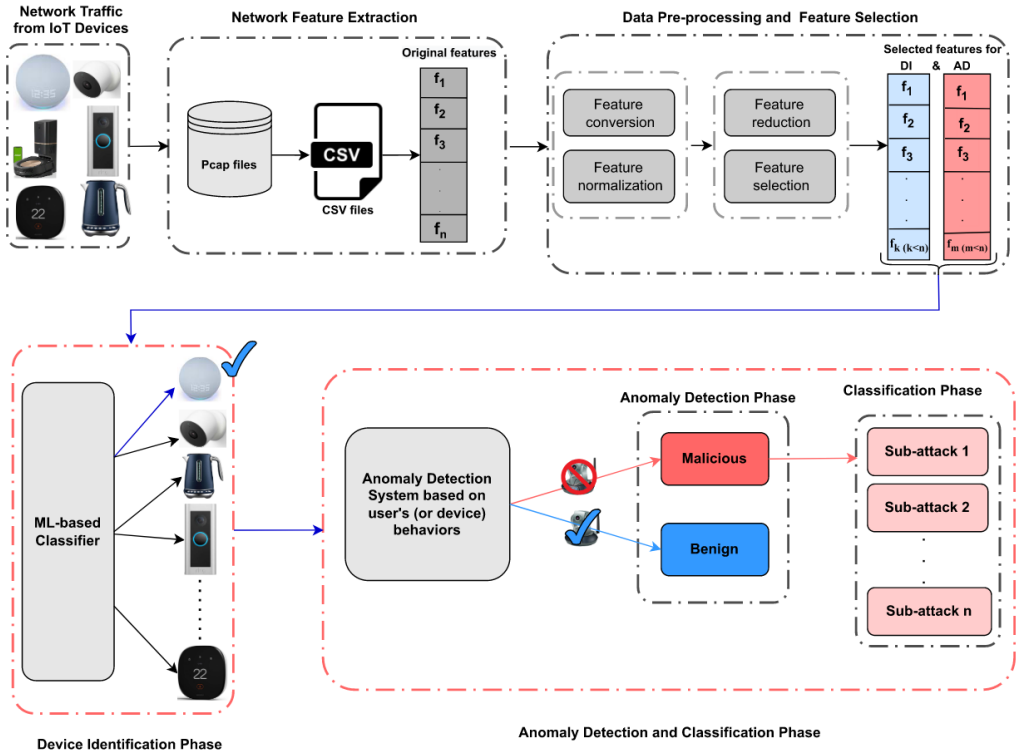

Figure 2.1: Workflow of the integrated framework for IoT device identification and anomaly detection (Rabbani et al., 2025).

- **Web-based attacks** — exploits directed at application-layer services on IoT devices, including injection attacks and cross-site scripting (XSS).
- **Brute-force** — repeated automated login attempts cycling through credential combinations to gain unauthorized access.
- **Spoofing** — falsification of IP, MAC, or protocol identifiers to impersonate legitimate devices and bypass authentication or monitoring.
- **Mirai** — malware that recruits compromised IoT devices into a botnet for coordinated large-scale attacks. Mirai botnets have been documented with as many as 400,000 devices available for hire (Kolias et al., 2017).

These attack categories represent a comprehensive range of network threats observed in IoT environments and are used to evaluate the performance of the hybrid framework, with the results of this evaluation summarised in Table 2.4.

Performance Metric Definitions:

- **Accuracy:** The proportion of total predictions that were correct, representing overall model performance.
- **Recall:** The proportion of actual attack events correctly detected by the model.

Table 2.4: Comparative results against recent anomaly detection and device identification techniques. Data adapted from (Rabbani et al., 2025).

Reference	DI	AD	Accuracy	Recall	Precision	F_1 Score	Dataset
Ruijie et al.	Yes	No	0.985	0.983	0.985	0.984	CICIoT2022
Bakhsh et al.	Yes	No	0.9642	0.9642	0.9642	0.9642	CICIoT2022
Zhao et al.	Yes	No	0.9874	0.9722	0.9868	0.9784	CICIoT2022
Behera et al.	No	Yes	0.9978	0.9876	1.00	0.9845	CICIoT2023
Jeffrey et al.	No	Yes	0.9319	0.9319	0.9353	0.9324	CICIoT2023
Zohourian et al.	No	Yes	0.9874	0.9971	0.9384	0.9529	CICIoT2023
Proposed DI approach	Yes	–	0.9999	0.9999	0.9999	0.9999	CICIoT2023
Proposed AD approach	–	Yes	0.9994	0.9991	0.9989	0.9990	CICIoT2023

- **Precision:** The proportion of detected attacks that were correctly classified, indicating how many alerts were accurate.
- **F1-score:** The mean of precision and recall, providing a balanced measure of the model's detection ability and its false-positive control.

Table 2.4 presents the comparative performance of Rabbani et al.'s (2025) hybrid device identification and anomaly detection framework against several earlier IoT traffic analysis techniques. Their proposed model achieved nearly perfect performance across all evaluated metrics. These results significantly outperform previous methods evaluated on the same dataset: *CICIot2023* (Canadian Institute for Cybersecurity (CIC), 2023). The findings demonstrate the effectiveness of integrating packet-based and flow-based approaches for comprehensive IoT traffic analysis.

2.5 Machine Learning for IoT Traffic Analysis

Both device identification and anomaly detection in IoT traffic begin with the same foundation: extracting and selecting the features that best describe device behaviour. This process includes normalizing and cleaning the data to ensure variables fall within expected ranges and that missing values are handled consistently (Neto et al., 2023; Yang et al., 2021). The quality of this preprocessing stage directly influences how well any subsequent classification model performs.

Machine learning has proven particularly effective at this task, improving both scalability and accuracy in device identification compared to rule-based approaches. Algorithms trained on extracted network features can automatically classify device types and detect behavioural anomalies through pattern recognition across large datasets (Yang et al., 2021).

Among the classifiers applied to IoT traffic analysis, Random Forest (RF) has emerged as a

consistently strong performer across both device identification and anomaly detection tasks. RF is a combined method that builds multiple decision trees during training and combines their outputs, improving generalization and reducing the overfitting risk that affects single-tree classifiers. Penna et al. (2025) found that while decision trees are a natural fit for threshold-based traffic classification, RF's resistance to overfitting makes it the more reliable choice, achieving 99% detection accuracy on real IoT network traffic. Safi et al. (2022) in their survey of profiling techniques, also identified RF as the best-performing classifier across the studies they reviewed. Rabbani et al. (2025) evaluated RF as part of their DIAD framework and found it to be competitive across both device identification and anomaly detection tasks.

Additionally, RF offers advantages that are relevant to IoT traffic analysis. Its training approach provides resilience when some device classes appear far more frequently in the data than others, which could be a common occurrence in heterogeneous IoT datasets. Within Python, Scikit-learn's RF module also provides a range of built-in tools, including feature importance scores, which would prove useful in understanding which traffic characteristics are most discriminative at each stage of a pipeline. These qualities make RF a well-supported choice for the kind of large-scale, multi-device classification tasks that IoT traffic analysis demands.

2.6 Key IoT Dataset for Evaluation

Rabbani et al. (2025) developed and evaluated their hybrid packet-flow machine learning framework using the CICIoT2023 dataset, which includes traffic from 105 IoT devices and 33 distinct attack types. Following this release, the dataset was further refined and expanded into the CIC IoT-DIAD 2024 version, designed specifically for hybrid device identification and anomaly detection research (Canadian Institute for Cybersecurity (CIC), 2024).

The dataset captures traffic from devices spanning seven functional categories: cameras, audio devices, smart plugs, home automation, lighting, hub devices, and sensors. Each under both benign and attack conditions. This range of device types reflects the heterogeneous nature of real IoT deployments discussed earlier in this chapter. Of the seven attack categories present in the dataset, several are particularly relevant to behavioural analysis. Mirai traffic is notable because it recruits devices gradually, meaning early-stage infection may produce only subtle deviations from benign behaviour. Reconnaissance activity similarly produces low-volume probing traffic that can be difficult to distinguish from normal background communication. DDoS and DoS attacks, by contrast, produce sharp and sustained spikes in flow volume and packet rate that are more immediately visible in flow-level metrics. These differences in how attacks manifest in traffic data make the dataset well-suited to evaluating whether behavioural features alone can support both device identification and anomaly detection across a realistic range of threat scenarios.

2.7 Summary

Clearly, no single detection approach is sufficient for the demands of a heterogeneous IoT environment. Signature-based systems like Snort are reliable against known threats but cannot adapt to novel or evolving attacks. Packet-based analysis can fingerprint individual devices with high accuracy, but captures only a snapshot of each transmission. Flow-based metrics reveal behavioural trends over time but lack the granularity to identify specific devices on their own. The hybrid approach addressed by Rabbani et al. (2025) demonstrates that combining both layers significantly improves performance across both tasks, but the challenge of scaling this to large and diverse device populations remains an open problem.

Machine learning, and RF in particular, has emerged from the literature as a well-suited tool for this kind of large model behavioural classification. Its consistent performance across IoT device identification and anomaly detection tasks, combined with its resilience to class imbalance, makes it a natural fit for evaluating the hybrid framework approach.

This study takes the challenge of a lack of a unified detection approach in IoT DIAD as its starting point. Using the CIC IoT-DIAD 2024 dataset as a test pool, it evaluates whether carefully selected behavioural features can support reliable device identification and anomaly detection across multiple IoT devices and a range of attack scenarios, without relying on static identifiers or payload inspection. The methodology used to pursue that evaluation is described in the following chapter.

3

Methodology

3.1 Introduction

This chapter explains how the study was conducted and why each method was chosen. The project uses a quantitative approach based on secondary data from the CIC IoT-DIAD 2024 (Canadian Institute for Cybersecurity (CIC), 2024) dataset, evaluating whether packet and flow-level behavioural features can support IoT device identification and anomaly detection. Device identification is evaluated across all 50 devices in the dataset, while anomaly detection is evaluated across three camera devices against six attack types selected to provide coverage across five attack categories. To achieve this, four research methods were used: a literature survey and review, dataset exploration, deductive reasoning, and a structured analytical approach using Random Forest classification. Each is described in the following sections.

3.2 Research Methodology

A research methodology is the structured framework that guides *how* a study is conducted. It defines the approach taken to gather, analyze, and interpret data in a way that is consistent with the research objectives (Bryman, 2016). In this study, a quantitative research methodology is used, which means the analysis is grounded in numerical data and measurable outcomes and not subjective interpretation. This is appropriate given that the primary objective involves evaluating the performance of classification models against defined metrics

such as accuracy, precision, recall, and F1-score.

The study draws on secondary data from the CIC IoT-DIAD 2024 dataset, which was captured under controlled conditions using real IoT devices. The focus is on analyzing existing labelled data to identify behavioural patterns that distinguish devices and reveal anomalous activity. This positions the study as exploratory in nature, as it is not testing a pre-formed hypothesis. The goal was instead to build and evaluate a framework through progressive experimentation. The specific methods used to carry this out are described in Sections 3.3 through 3.6, followed by a discussion of the primary dataset, evaluation metrics, and limitations of the approach.

3.3 Literature Survey and Review

A literature survey involves collecting and studying relevant academic papers, reports and datasets to gain an understanding of the current state of research (Carrera-Rivera et al., 2022). In this study, the literature review serves three purposes:

1. To understand how IoT devices behave on networks and how their traffic can be profiled
2. To examine existing methods of device identification and anomaly detection, with particular focus on packet-based and flow-based techniques
3. To establish the theoretical foundation for the analytical framework developed and evaluated in this study

Three academic databases were consulted during the review, summarized in Table 3.1, using the search parameters listed in Table 3.2. Publications were selected based on relevance to IoT traffic analysis, device profiling, and anomaly detection, with preference given to studies published within the last five years.

Table 3.1: Databases explored

No.	Database Name	Address
1	ACM Digital Library	https://dl.acm.org/
2	ScienceDirect	https://www.sciencedirect.com/
3	ResearchGate	https://researchgate.net/

Table 3.2: Search parameters

No.	Search Parameter
1	“IoT device identification”
2	“IoT device fingerprinting”
3	“packet-based IoT analysis”
4	“flow-based anomaly detection”
5	“IoT network attacks”
6	“IoT traffic analysis”
7	“IoT behaviour profiling”

3.4 Dataset Exploration and Review

Before any ML classification models were trained, the CIC IoT-DIAD 2024 dataset was explored to understand its structure, traffic features and attack categories. This exploration was needed to make informed decisions about feature selection and how to partition the data for training and testing.

During exploration, the following aspects were examined:

1. The fifty unique IoT devices present in the dataset and their functional categories
2. The structure and distribution of benign traffic per device
3. The metadata, timing, and flow-level metrics specific to each device
4. The structure and labelling of malicious traffic across all seven attack categories

This stage provided the foundation for understanding how each device communicates under normal, benign conditions, which is a needed prerequisite for anomaly detection. The primary dataset used throughout this study is described in detail in Section 3.7.

3.5 Deductive Reasoning and Pattern Identification

Deductive reasoning involves beginning with an established theory and testing whether observed data confirms or contradicts it (Bryman, 2016). In this study, deductive reasoning connects theoretical expectations drawn from the literature to the patterns observed in the CIC IoT-DIAD 2024 dataset. For example, network security theory suggests that a DoS attack generates abnormally high packet rates compared to benign behaviour. This expectation can be tested by analyzing the dataset’s traffic flows and determining whether the observed attack patterns match the predicted deviation from normal behaviour.

This approach is applied throughout both the device identification and anomaly detection stages of the study. In device identification, the expectation is that devices from different functional categories will produce behaviourally distinct traffic patterns. In anomaly detection, the expectation is that attack traffic will deviate measurably from the benign baseline established for each device. These theoretical expectations guided both the feature selection process and the design of the classification framework, which is described in the following section.

3.6 Proposed Framework Development and Evaluation

This section describes the analytical framework developed and evaluated in this study, referred to throughout as the Device Identification and Anomaly Detection (DIAD) pipeline. The framework follows the hybrid packet- and flow-based approach established in the literature, applying Random Forest classification at each stage.[1] Random Forest was selected due to its consistently strong performance in IoT device identification and anomaly detection tasks, its robustness to class imbalance, and its ability to handle high-dimensional feature spaces without extensive hyperparameter tuning (Neto et al., 2023; Safi et al., 2022). All experiments use an 80/20 stratified train-test split to preserve class distributions during evaluation.

Reproductibility note: A fixed random seed of 42 is applied to all components of the pipeline (the train/test split, the Random Forest bootstrap sampling, and per-device row cap in DI Model 3). This ensures that reported results are reproducible from the same input data, and that differences between models reflect the models themselves and not variance in the random splits.

Figure 3.1 presents the proposed DIAD pipeline as evaluated in this study. The framework begins with feature extraction from the FlowMeter refined dataset, where packet- and flow-level behavioural features are normalized and stripped of static identifiers to ensure classification relies on communication behaviour alone.

Device identification is then evaluated through three progressive classification models, moving from broad device categories through per-category instance identification to full fifty-device classification. A reduced set of nine features is then selected for the anomaly detection stage, made up of stream, channel and jitter statistics and summaries across five temporal aggregation windows. These features then feed into two anomaly detection tasks. The first is a binary classification of the traffic being benign or anomalous. and the second as a multi-class classification of anomalous traffic into specific attack types. The anomaly detection stage is evaluated on three camera devices (Arlo Q Indoor, Nest Indoor and Netatmo Camera) against six attack types selected to provide coverage across five attack categories.

At a high level, the approach proceeds as follows:

[1] The full implementation of all five models, along with the DIAD pipeline diagram, is provided in Appendix G.

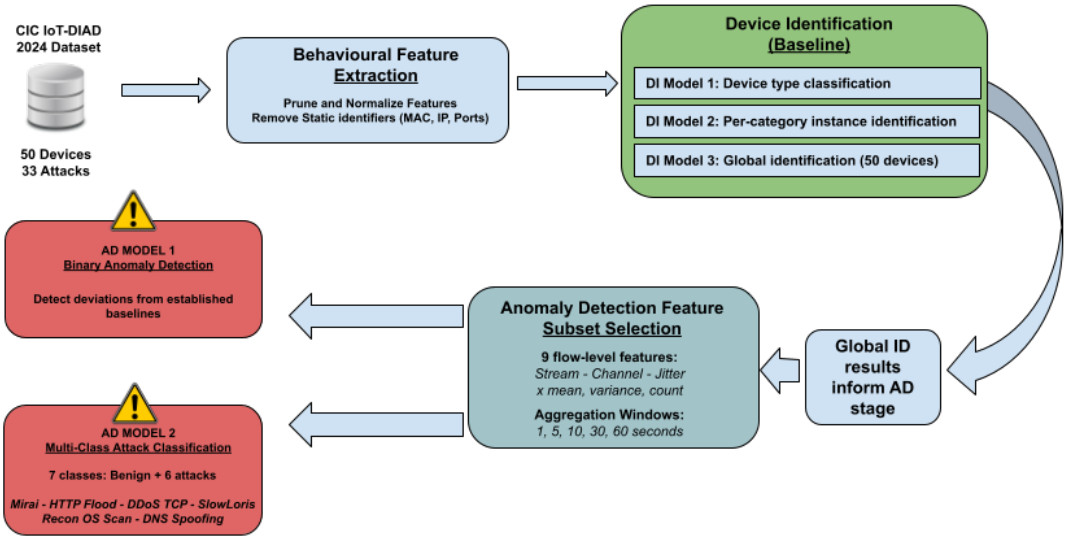

Figure 3.1: Proposed DIAD framework showing the specific experimental configuration evaluated in this study.

- **Device identification (packet-level):** Packet headers, protocol fields, TLS/HTTP metadata, payload size, and related features are exported and used to identify and build behavioural fingerprints for each IoT device during known benign periods.
- **Normal baseline construction (flow-level):** *Stream, channel, and jitter metrics* are isolated per device to characterize benign timing, volume, and communication patterns.
- **Anomaly detection:** Malicious traffic data is compared against these baselines; significant deviations in behaviour are flagged as potential anomalies and categorized by attack type.
- **Dataset comparison and validation:** Flagged anomalies are then compared with the attack labels in CIC IoT-DIAD 2024 to evaluate how well the approach correctly identifies devices, and its success in distinguishing benign from malicious activity.

3.6.1 Data Exploration and Preprocessing

The CIC IoT-DIAD 2024 dataset provides packet-based feature files extracted from raw packet capture (pcap) files. Packet-based files contain 133 features organized into five categories (Rabbani, 2024):

1. **Network Protocol Features** - These include: TCP/UDP/HTTP protocols, TTL, ethernet frame size, packet size, TCP window size, payload length.
2. **Handshake and TLS Features** - These include: TLS Handshake version, length of signature and hash algorithms in TLS handshake.
3. **Timing Features** - These include: Inter-arrival time, time since the previously displayed frame.

4. **Statistical Features** - Metrics that summarize packet behaviour such as mean, median, variance, minima, maxima, and packet counts.

5. **User Agent Features** - These include: User Agent browser, User Agent OS, User Agent device.

In addition to these descriptors, the packet-based files include flow-aggregated stream, channel and jitter statistics computed across five temporal windows (1, 5, 10, 30, and 60 seconds). These form the basis of the anomaly detection feature set described in Section 3.6.3.

Prior to training, static identifiers, including MAC addresses, IP addresses, port numbers, vendor strings, application layer strings, and non-numerical columns were removed from the feature set. This ensures that classification models rely exclusively on communication behaviour and not spoofable static attributes, consistent with prior work demonstrating that behavioural features provide more robust and generalizable device fingerprints (Okey et al., 2025; Xu et al., 2025).

3.6.2 Device Identification Classification Models

Device identification is evaluated through three classification models of increasing complexity. The first model classifies traffic into broad functional device categories. The second performs instance-level identification within each category, training a separate Random Forest model per category. The third performs global multiclass identification across *all* devices simultaneously, representing the most demanding configuration and the closest approximation to real-world deployment conditions. Together, these three models evaluate how reliably behavioural baselines can be established at different levels of granularity, from broad device type down to individual device instance.

Note that the *Sensor* device category only contained a single labelled instance in the dataset and was therefore excluded from per-category instance-level identification in DI Model 2.

3.6.3 Anomaly Detection Classification Models

Anomaly detection is evaluated across three camera devices: the Arlo Q Indoor Camera, the Nest Indoor Camera, and the Netatmo Camera. Camera devices were selected due to their position as the highest-traffic category in the dataset, providing sufficient volumes of both benign and attack-labelled samples for meaningful evaluation across multiple devices.

The cameras are evaluated against six attack types: Mirai, DoS HTTP Flood, DDoS TCP Flood, DDoS SlowLoris, Recon OS Scan, and DNS Spoofing. Selection was based on two criteria: sufficient sample volume across all three devices and coverage across attack categories. One attack was selected from each category where sufficient data existed, with two DDoS variants included to compare high-intensity flood behaviour across different protocols. BruteForce was excluded as it returned zero rows for all three devices. Web-Based attacks

were also excluded due to insufficient traffic volume. The available attack traffic per device across all 26 attack types is presented in Appendix E.

Balanced datasets were constructed for each device and attack combination by extracting benign traffic from the devices' recorded baseline and pairing it with the corresponding attack traffic, undersampling the majority class to maintain equal class distribution. This resulted in 18 balanced CSV files, one per device and attack combination, and they were used as inputs to the classification models.

Two classification tasks are evaluated. The first is binary anomaly detection, determining whether observed traffic is benign or anomalous. The second is multi-class attack classification, identifying the specific attack type present. Both tasks use a compact set of nine flow-level features derived from stream, channel, and jitter statistics, summarized in Table 3.3. Traffic is aggregated across multiple time windows of 1, 5, 10, 30, and 60 seconds to evaluate the effect of aggregation duration on classification performance. This intentionally reduced feature set reflects findings in the literature that carefully selected behavioural features can maintain strong detection performance while reducing computational overhead (Okey et al., 2025).

Table 3.3: Nine flow-level features used for anomaly detection, where `{w}` denotes the aggregation window size (1, 5, 10, 30, or 60 seconds).

Metric	Feature	Description
Stream	`stream_{w}_count`	Number of stream flows in window
Stream	`stream_{w}_mean`	Mean stream flow size in window
Stream	`stream_{w}_var`	Variance of stream flow size in window
Channel	`channel_{w}_count`	Number of channel flows in window
Channel	`channel_{w}_mean`	Mean channel flow size in window
Channel	`channel_{w}_var`	Variance of channel flow size in window
Jitter	`stream_jitter_{w}_sum`	Total jitter across stream in window
Jitter	`stream_jitter_{w}_mean`	Mean jitter across stream in window
Jitter	`stream_jitter_{w}_var`	Variance of jitter across stream in window

3.6.4 Implementation Tools and Libraries

All experiments were implemented in Python, using `scikit-learn` for all machine learning components, including `RandomForestClassifier` for classification, `train_test_split` with stratification for data partitioning, with `classification_report` and `confusion_matrix` for evaluation. `SimpleImputer` and `Pipeline` were used to ensure consistent preprocessing between training and test data, and `joblib` was used to save trained models to disk to allow results to be reproduced without retraining. `Pandas` and `NumPy` handled data loading, feature selection and array operations throughout. Confusion matrix figures were generated using `Matplotlib`.

3.7 Primary Data Set

The primary dataset used in this study is the CIC IoT-DIAD 2024, developed by the Canadian Institute for Cybersecurity (Canadian Institute for Cybersecurity (CIC), 2024). It captures network traffic from 105 real IoT devices across 33 distinct attack types, classified into seven categories: DDoS, DoS, Reconnaissance, Web-based, Brute Force, Spoofing, and Mirai. All attacks are executed by malicious IoT devices targeting other IoT devices within the same network topology (Rabbani et al., 2025). A full list of devices included in the dataset is provided in Appendix A.

3.8 Evaluation Metrics

To evaluate the performance of the classification models at each stage of the DIAD pipeline, this study uses four standard metrics derived from the confusion matrix (Amara Korba et al., 2025). For each metric, TP denotes true positives, TN true negatives, FP false positives, and FN false negatives.

Accuracy measures the overall proportion of correct identifications:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures how many of the flows identified as attacks are actually attacks:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures how many of the real attacks the model successfully identifies:

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-score provides a balanced measure combining precision and recall:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

3.9 Limitations

This study uses a comprehensive dataset, but several limitations must be recognized:

- The CIC IoT-DIAD 2024 dataset contains a predefined set of 33 attack types. Models trained on this data are therefore restricted to these specific attacks, and any real-world threats not represented in the dataset, including zero-day exploits, will not be directly

identifiable. However, a behaviour-based detection system may still flag deviations from normal traffic as suspicious, even if the specific attack type cannot be classified.

- The dataset contains 50 unique devices; the anomaly detection evaluation is limited to three camera devices, and six attacks as a proof of concept. While this allows for detailed per-device behavioural analysis, the findings may not generalize directly to all device types without further evaluation.

- Although the dataset was captured using real IoT devices, they were operated under controlled and simulated network conditions. This may not fully reflect the complexity of live network environments, including device misconfigurations, firmware variations, or mixed vendor deployments, which could affect framework performance in real-world settings.

- Hyperparameters were not tuned. Default values were used based on Random Forest's documented robustness to parameter variation in classification.

- In the current setup, the device identification models run sequentially: Model 2 inherits any misclassifications from Model 1, since a device assigned to the wrong category cannot be correctly identified by the corresponding per-category classifier. Given DI Model 1's accuracy of 99.70%, this compounding error is small but non-zero.

- While training time is not a focus of this study, the processing power of the hardware available was limited. Future research conducted on more capable hardware may be able to extend the scope of experimentation, for example, by evaluating anomaly detection across a wider range of devices rather than this study's three devices and six attacks.

3.10 Chapter Summary

This chapter described the methodology behind the study's behavioural detection framework. Working from the CIC IoT-DIAD 2024 dataset, the proposed DIAD pipeline combines packet-level device identification with flow-level anomaly detection, using Random Forest classifiers at each stage. Device identification is evaluated through three progressive models of increasing granularity, and anomaly detection across three camera devices and six attack types using a compact nine-feature flow representation. With the approach and its limitations presented, Chapter 4 presents the results of the DIAD framework evaluation, addressing each of the research objectives while working through each stage of the framework from device identification to anomaly detection.

4

Results/Analysis

4.1 Chapter Overview

This chapter presents the results from training multiple machine learning models on the CIC IoT-DIAD 2024 dataset for the purpose of IoT device identification (DI) and anomaly detection (AD). DI results are presented first and are used to evaluate whether packet-based features capture sufficiently stable behavioural characteristics to distinguish devices under benign operating conditions. This stage sets a baseline of normal device behaviour, which is required before meaningful AD can be performed.

To achieve this, DI is examined through a series of three classification models with increasing complexity. These models progressively evaluate ID at the device type level, device categories, individual devices and across the full dataset. The purpose of this progression is to determine how reliably behavioural baselines can be established at different levels of granularity.

Following this, AD results are presented using strict flow-based traffic features. This stage represents the primary security-focused component of the study and examines how network behaviour deviates from benign baselines under different attack scenarios. Flow-level metrics are analyzed to determine their effectiveness in revealing anomalous activity across three camera devices and six attack types, and whether these metrics can also be judged to correctly pair a device with a specific attack. The chapter concludes with a discussion of the findings across both stages and an implementation and evaluation of the two framework stages of the DIAD framework that was developed from this process.

4.2 Dataset and Experimental Setup

This study uses the CIC IoT DIAD 2024 dataset, containing labelled IoT traffic using both benign and multiple attack scenarios. The dataset includes diverse IoT device classifications ranging from cameras and audio devices to smart lighting and smart plugs. For the device DI experiments, analysis is restricted to benign traffic only, in order to prevent attack-related behaviours from influencing device fingerprinting. Feature extraction is based on the packet-based behavioural feature set provided in the CIC IoT-DIAD 2024 dataset, which includes both packet-level descriptors and flow-aggregated stream, channel, and jitter statistics. Static identifiers such as MAC addresses, IP addresses, port numbers, vendor information, and application layer strings are removed to prevent identity leakage, consistent with prior work demonstrating that IoT devices should be characterized through communication behaviour instead of through static identifiers (Okey et al., 2025). This *behavioural-only* selection method ensures that classification models rely only on communication patterns.

All experiments are performed using Random Forest (RF) classifiers. This model is selected due to its strong performance in IoT device identification tasks. Prior studies report that RF consistently outperforms alternative classifiers in behavioural traffic-based IoT profiling scenarios (Safi et al., 2022). As for AD, Random Forest has demonstrated stable performance across binary and multiclass IoT attack detection tasks while evaluating similar datasets (Neto et al., 2023).

Within each model configuration, data is partitioned using an 80/20 train-test split to preserve class distributions during evaluation. To address class imbalance and prevent high-volume (cameras) from disproportionately guiding model learning, balancing strategies such as per-device sample caps are applied where appropriate. These mechanisms ensure stable and fair evaluation across device classes while maintaining realistic classification difficulty.

Results are organized into two stages: *device identification*, which evaluates the ability of packet and flow-level features to distinguish devices under benign conditions, and *anomaly detection*, which evaluates the ability of flow-level features to detect and classify attack traffic. In each stage, models are presented in order of increasing complexity, with a summary provided at the end of each stage before moving to the next.

4.3 Device Identification Models and Results

Device identification is evaluated through three progressive classification models. Each model increases in complexity and scope, moving from broad device category classification down to full instance-level identification across all 50 devices simultaneously. This progression is designed to establish how reliably behavioural baselines can be constructed at different levels of granularity, and to identify where the limits of behavioural separability begin to appear.

4.3.1 DI Model 1: Device Type Classification

(What type of device is this?)

The first device identification model addresses the broadest classification task: "Given a set of behavioural traffic features, can the model correctly determine what functional category a device belongs to?"

This serves as an initial validation step, establishing whether category-level behavioural patterns exist before attempting finer-grained instance-level identification. Traffic is classified into broad functional categories: Cameras, Audio Devices, Power Outlets, Home Automation, Lighting, Hub Devices and one Sensor. Beyond this model, the single Sensor was not included in further classification comparisons, as it lacked multiple devices in the same category for classification models.

Table 4.1: Device Type Identification Performance (Behavioural Network Features)

Device Type	Precision	Recall	F1-score	Support (80/20)
Audio Device	0.9988	0.9994	0.9991	24,064
Camera	0.9996	0.9998	0.9997	75,801
Home Automation	0.9969	0.9884	0.9926	3,883
Hub Device	0.9987	0.9957	0.9972	6,240
Lighting	0.9972	0.8942	0.9429	2,392
Power Outlets	0.9725	0.9988	0.9855	10,030
Sensor	0.9962	0.9962	0.9962	786
Overall Accuracy			0.9970	
Macro Avg F1			0.9876	
Weighted Avg F1			0.9969	

The first device ID experiment evaluates whether packet-based features provide enough resolution to tell the difference between broad IoT device categories under benign operating conditions. This experiment serves as an initial validation step and shows whether category-level behavioural patterns exist to support later baseline construction. The dataset contained 615,980 benign flows. An 80/20 stratified split yielded 123,196 test samples across seven device categories. The model achieved an overall classification accuracy of (99.7%), indicating strong identifiability between device types. The high macro F1 (98.76%) confirms that minority categories were not disproportionately misclassified, while the close alignment between weighted F1 (99.69%) and overall accuracy (99.7%) indicates stable performance across dominant classes.

Most device categories exhibited near-perfect separability. Cameras (F1 = 99.97%) and Audio Devices (F1 = 99.91%) demonstrated highly distinctive behavioural signatures. Camera traffic typically includes sustained bidirectional flows and consistent packet-size distributions associated with streaming and cloud synchronization, while smart audio devices' behaviour would be dependent on structured command–response exchanges. Ultimately, the two busiest device categories' behavioural differences appear sufficient for reliable classification. The

only notable reduction in performance occurred within the Lighting category (F1 = 94.29%), primarily due to reduced recall (89.42%). Precision remained high (99.72%), indicating mostly correct classification. This pattern could be showing behavioural similarity between Lighting devices and other low-complexity systems, such as Power Outlets (98.55%). Short command bursts would most likely reduce behavioural uniqueness at this high granularity level.

These results confirm that packet-level behaviour can display clear differences between device categories. This supplies a validated baseline for higher-resolution identification models.

4.3.2 DI Model 2: Device Identification within Device Type

How accurately individual devices are identified within each device category[1]

The second device identification model narrows the classification task: "Given devices from the same functional category, can the model correctly identify which specific device is communicating?"

Instead of attempting a 50-class global classification, separate Random Forest models were trained *per category* to determine how accurately individual devices could be distinguished among behaviourally similar peers. Individual devices are therefore labelled and distinguished from others belonging to the same device type. This model focuses on in-category separability and removes cross-category variance as a confusion factor. Each model was trained using the same strict feature set established for Model 1, with identity-leaking features removed. Table 4.2 summarizes the in-category instance identification performance.

Table 4.2: Instance-Level Device Identification Performance (Per-Category Multi-Class Classification)

Category	Devices	Samples (Test)	Accuracy	Macro F1
Camera	14	75,802	0.9979	0.9853
Audio Device	7	24,065	0.9953	0.9936
Home Automation	6	3,883	0.9992	0.9981
Hub Device	4	6,240	0.9399	0.9330
Lighting	4	2,393	0.9983	0.9985
Power Outlets	14	10,030	0.8755	0.8978

In-category performance remained high for most groups. Cameras, which represent by far the busiest of all the device groups, achieved 99.79% accuracy, indicating that even a high volume of similar traffic can be correctly identified. Audio devices and Home Automation systems similarly demonstrated strong separability, with macro F1 values above 99%. Lighting devices, despite containing the lowest number of testing samples, achieved a 99.83% accuracy and macro F1 of 99.85%. This contrasts with the category-level recall drop observed in

[1] DI Model 2 Full results found in Appendix B

Model 1, suggesting that once isolated from other low-flow devices, different lighting classes become more behaviourally distinct.

The most significant reduction in performance occurred within the Power category (accuracy = 87.55%, macro F1 = 89.78%). This category contained 14 device instances, many of which share similar low-throughput command–response behaviour, typically performing simple "on/off" actuation. Several devices originate from the same manufacturer families (e.g., multiple GoSund variants, paired Teckin, Wemo, and Yutron plugs), resulting in highly comparable traffic patterns. These devices generate low-volume, event-driven communication with similar packet size distributions and timing characteristics, leading to overlapping behavioural signatures and reduced separability at the instance level. This decline in performance could probably be viewed as a reflection of similarity in device behaviour and not weakness in the model.

4.3.3 DI Model 3: Full Device Identification Across All Devices

Which exact device is this across the entire dataset?[2]

The third and most demanding device identification model removes all category boundaries: "Across all 50 devices in the dataset, can the model correctly identify which specific device is communicating using behavioural features alone?"

While Model 2 evaluates separability within individual device categories, the next stage increases the complexity of the task by removing category boundaries. Model 3 performs global multiclass identification across all device instances simultaneously, requiring the classifier to distinguish between *all* device instances. This configuration builds the most comprehensive model of behavioural fingerprinting performance under realistic deployment conditions.

A Random Forest classifier is trained to perform full multiclass device identification using 83 behavioural features following the removal of identity-leaking attributes (the packet based feature set after removal of static and non-numerical columns, as described in Section 3.6.1. Although this represents the complete behavioural feature set retained after preprocessing, discriminative performance may be driven by a smaller subset of highly informative features, as suggested by feature importance analysis.

To maintain stability under this larger configuration, a memory-friendly training setup is applied. Per-device sample caps are introduced (only affecting Cameras), tree depth and minimum leaf size are controlled, and each tree is trained on a subsampled portion of the dataset. These adjustments significantly reduce computational load while preserving strong identification performance, and are referred to in Table 4.4 as the low-memory variant. The hardware constraints that motivated these decisions are discussed in Section 3.9. The reported

[2] DI Model 3 Full results found in Appendix C

Model 3 results were produced with the script's default configuration (`--max-per-device 100000`, `--n-estimators 300`, `--max-samples 0.8`); these are exposed as command-line flags and can be reduced for memory-constrained hardware.

Table 4.3: Overall performance for IoT device identification (Random Forest, 50 devices, max 100,000 samples per device).

Metric	Value
Number of devices (classes)	50
Feature count	83
Training set size	376,258
Test set size	94,065
Accuracy	0.9532
Macro-average F1-score	0.9019
Weighted-average F1-score	0.9534

As mentioned previously, all device instances are evaluated within a single multiclass setting, meaning the classifier must distinguish between all fifty devices at once. This is the most challenging configuration tested. Even under this broader label-space, the model maintains strong overall performance (accuracy = 95.32%, weighted F1 = 95.34% see Table 4.3), showing that packet-level behavioural features remain a strong discriminator when identification is scaled up. Although performance decreases slightly relative to the earlier models, this reduction is expected, given that devices from different categories and manufacturers must now be separated within a single classification task. Notably, the decline is gradual, not sudden, suggesting that behavioural fingerprinting scales predictably as identification complexity increases.

Table 4.4: Comparative summary of device identification (DI) models

Feature / Factor	Model 1	Model 2	Model 3
Objective	Device type prediction	Instance ID within category	Instance ID across all devices
Classes	7 categories	4–14 per category	50 devices
Feature Count	83	83	83
Dataset Scope	All devices	Per-category subsets	All devices
Row Capping	No	No	Yes (low-memory variant)
Tree Constraints	300 trees	300 trees	Reduced depth + subsampling
Accuracy	99.70%	87.55%–99.92%	95.32%
Macro F1	98.76%	89.78%–99.85%	90.19%
Hardest Case	Lighting	GoSund plug family	GoSund plug family

4.3.4 DI Results Summary

Table 4.4 summarizes the progression of device identification models evaluated in this study, and compares their configuration, scope, and performance across key metrics.

The three configurations increase in classification granularity and complexity. Model 1 operates at the functional level, distinguishing device categories and establishing that behavioural features alone provide strong separability between device families. Model 2 increases resolution by performing instance-level identification within each category, isolating behavioural differences among structurally similar devices. Models 1 and 2 were trained using an identical strict behavioural feature set (85 numeric features), ensuring that performance differences reflect classification resolution instead of feature variation. While most categories remain highly separable, performance degradation is observed among homogeneous device families such as power outlets, showing difficulty in separating some behavioural overlap.

Model 3 represents the most demanding configuration, performing global multiclass identification across all 50 devices simultaneously. This introduces cross-category competition and provides an outlook for potential successful scalability. Even under controlled training conditions designed to reduce computational overhead (low memory errors), global identification keeps a high level of accuracy, demonstrating that behavioural fingerprinting remains strong in resource-limited operating scenarios.

The comparisons show a clear trade-off between classification granularity and performance. As model scope expands from category-level to full instance-level, macro-level performance decreases slightly, but remains strong. This progression shows the stability of behavioural features under benign conditions and sets a reliable foundation for further anomaly detection experiments. The following section presents the anomaly detection stage of the DIAD pipeline, beginning with binary classification before moving to multi-class attack identification.

4.4 Anomaly Detection Models and Results

This section evaluates the anomaly detection stage of the DIAD pipeline, extending the evaluation across three camera devices: the Arlo Q Indoor Camera, the Nest Indoor Camera, and the Netatmo Camera, against six attack types selected to provide coverage across five attack categories. Camera devices were selected due to their position as the highest-traffic category in the dataset, providing sufficient volume of both benign and attack-labelled samples for meaningful evaluation across multiple devices.

Unlike the full feature pipeline employed by Rabbani et al. (2025), this study deliberately reduces the anomaly detection feature set to nine flow-aggregated features derived from stream, channel, and jitter statistics. These are then summarized as mean, variance, and count. This reduction is intentional; by not replicating an established high-performing pipeline,

the goal here is to evaluate how much detection capability can be preserved using a minimal, lightweight feature representation.

Two classification tasks are evaluated, presented in the following subsections.

4.4.1 AD Model 1: Binary Anomaly Detection

Is this traffic anomalous? ***Yes*** *or* ***No***[3]

The first anomaly detection model is used to evaluate the simplest classification task: "*Given a set of flow-level behavioural features, can the model correctly determine whether observed traffic is benign or anomalous?*"

A Random Forest classifier is trained per device and attack combination using the nine behavioural features described in Table 3.3, evaluated across five temporal aggregation windows of 1, 5, 10, 30 and 60 seconds. Full per-window results for all three devices and attack combinations are provided in Appendix D. The key findings are found below.

Window Size Effect

Across all three devices and all six attacks, the 1-second window consistently produced the weakest detection performance. Performance improved steadily with longer aggregation windows, with most device and attack combinations stabilizing at or above 99% accuracy by the 10 to 30-second window. This pattern shows that moderate temporal aggregation provides sufficient behavioural context for reliable anomaly detection at a binary level. Notably, this improvement trend was not uniform across attack types. Mirai and HTTP Flood reached strong performance relatively quickly, while SlowLoris and Reconnaissance OS Scan showed a more gradual improvement curve, only reaching peak performance at longer windows. This suggests that the optimal aggregation window for binary detection is attack-dependent and not universal, with attacks that produce subtle or low-volume traffic deviations requiring longer windows to accumulate sufficient behavioural patterns.

Device-Level Comparison

The Arlo Q and Nest Indoor cameras demonstrated strong performance across all six attacks, with most combinations reaching 99% accuracy or above at the 30-second window, as summarized in Table 4.5.

[3] Results for AD Model 1 found in Appendix D

Table 4.5: Binary anomaly detection accuracy at the 30-second window across all devices and attacks.

Attack	Arlo Q	Nest Indoor	Netatmo
Mirai	0.9947	1.0000	0.9888
HTTP Flood	0.9912	1.0000	0.9734
DDoS TCP	1.0000	0.9975	0.9825*
SlowLoris	0.9987	0.9952	0.9847
Recon OS Scan	0.9980	0.9970	0.9592
DNS Spoofing	0.9924	0.9997	0.9827

* Statistically limited — 57 test samples only.

The Netatmo Camera produced slightly weaker results across all attacks, with several combinations failing to reach 98% accuracy even at the 60-second window. This gap correlates directly with Netatmo's lower benign traffic volume (10,844 rows compared to 143,546 for Nest Indoor). A richer benign baseline provides the model with a more complete characterization of normal device behaviour, improving its ability to detect deviations. This finding had implications for deployment: *devices with low traffic volume may require longer baseline capture periods before anomaly detection can be applied reliably*.

Attack-Level Observations

Recon OS Scan produced the weakest 1-second window results across all three devices, with Nest Indoor reaching only 80.56% accuracy at this window. This could be explained by the nature of reconnaissance traffic, which is designed to blend with normal background communication and produces only subtle deviations from benign behaviour at short aggregation intervals. DNS Spoofing and SlowLoris showed similar patterns, with weak 1-second performance improving dramatically at longer windows, suggesting that the temporal signature of these attacks only becomes detectable over sustained observation periods.

The Netatmo DDoS TCP results are noted as statistically limited, with only 57 test samples available due to low attack traffic volume for this device. These results should be interpreted with caution and are included for completeness and not as a reliable performance indicator.

Overall, binary anomaly detection proves reliable across all three devices and all six attack types at moderate aggregation windows. Moderate aggregation windows of 10 seconds or greater appear to offer the most stable performance, though the optimal window duration appears to be attack-dependent. The compact nine-feature representation is sufficient for strong binary detection performance, supporting the case for lightweight deployment in resource-limited IoT environments.

The following section extends this task to multi-class attack classification, evaluating whether the same compact feature set can distinguish between specific attack types.

4.4.2 AD Model 2: Multi-Class Attack Classification

If anomalous, which attack type is it?[4]

The second anomaly detection model extends the task from binary classification to multi-class attack identification: Given anomalous traffic from the devices, can the model correctly identify which specific attack type is present?

Instead of only determining whether traffic is benign or anomalous, the objective here is to distinguish between specific attack types. Again, using the Arlo Q, the Nest Indoor, and the Netatmo cameras as a case study, benign traffic is evaluated alongside six attack types selected to provide coverage across the CIC IoT-DIAD 2024 dataset's categories. One attack was selected from each category where sufficient traffic volume existed across all three devices: Mirai (BotNet), HTTP Flood (DoS), DDoS TCP (DDoS), SlowLoris (also DDoS, retained as a second high-intensity variant for comparison), Recon OS Scan (Reconnaissance), and DNS Spoofing (Spoofing). The BruteForce category was excluded, as it returned zero traffic rows for all three camera devices, and the Web-Based category was excluded due to insufficient volume across the devices evaluated. Combined with benign traffic, this produces a seven-class classification task.

The same nine features used in AD Model 1 are retained to allow controlled comparison. For each aggregation window, stream, channel and jitter statistics are summarized using mean, variance and count, giving a consistent nine-feature representation. The classifier then operates solely on temporal traffic dynamics. Balanced datasets were constructed per device by undersampling all seven classes to match the smallest class available for that device. This resulted in 566 samples per class for Arlo Q (constrained by HTTP Flood), 993 for Nest Indoor (constrained by DDoS TCP), and 142 for Netatmo (constrained by DDoS TCP). These differences in training volume reflect the underlying traffic availability per device and are an important factor in interpreting the results below.

Window Size Effect Across Devices

Performance across all aggregation windows is summarized in Table 4.6. At a 1-second window, overall accuracy is poor, dropping to as low as 61% for the Netatmo Camera, indicating that minimal aggregation does not provide sufficient context for reliable attack differentiation. Performance improves substantially at 5 and 10 seconds and reaches its highest value at the 60-second window. This behaviour differs from the binary detection results, where the optimal window size appeared to be attack dependent. At the multi-class level, longer aggregation windows appear to provide more stable behavioural signatures for discriminating between multiple attack types simultaneously. This is a notable finding, suggesting that distinguishing between seven classes rather than two requires a richer temporal context, and

[4] Full results for AD Model 2 found in Appendix E and additional confusion matrices for alternate devices found in Appendix F

the compact nine-feature representation benefits from longer observation periods when the classification task is more demanding.

Table 4.6: Multi-class classification performance across all three camera devices and five aggregation windows. The 60-second window achieved the best performance for all devices and is therefore used for the detailed analysis that follows.

Window	Arlo Q		Nest Indoor		Netatmo	
Size	Acc	F1	Acc	F1	Acc	F1
1s	0.7074	0.7062	0.7038	0.7044	0.6181	0.6095
5s	0.8600	0.8600	0.8483	0.8481	0.8191	0.8178
10s	0.9042	0.9040	0.8958	0.8957	0.7940	0.7930
30s	0.9357	0.9357	0.9468	0.9466	0.8492	0.8475
60s	**0.9470**	**0.9473**	**0.9626**	**0.9626**	**0.8593**	**0.8579**

Device-Level Comparison

In the 60-second window (Table 4.6), the Nest Indoor Camera achieved the strongest performance (96.26% accuracy, 96.26% macro F1), followed by the Arlo Q Indoor Camera (94.70% accuracy, 94.73% macro F1). The Netatmo Camera produced the weakest result (85.93% accuracy, 85.79% macro F1). This ordering correlates directly with the amount of benign baseline traffic available per device: Nest Indoor trained on 993 samples per class, Arlo Q on 566, and Netatmo on only 142. The roughly ten-point accuracy gap between Nest Indoor and Netatmo shows the finding from the binary detection results, that devices with richer benign baselines support stronger deviation-detection performance. As the task of classification is more focused, the broader principle remains for AD: *the baseline quality is a key factor in behavioural anomaly detection reliability*.

Detailed Analysis: Arlo Q Indoor Camera

The Arlo Q Indoor Camera is featured as the primary focus for detailed class-level analysis, as it sits in the middle of the three devices in terms of benign baseline volume. Class-level performance at the 60-second window is presented in Table 4.7. All seven classes achieve F1-scores above 92%, with DDos TCP showing the strongest performance (F1= 98.21%) and Recon OS Scan the weakest (F1=92.86%). Precision and recall are well-balanced across most classes, with DNS Spoofing showing an inverse pattern of high recall (98.25%) but lower precision (88.19%). This indicates that the classifier occasionally mislabels traffic from other classes as DNS Spoofing.

The confusion matrix in Figure 4.1 shows a more detailed view of where errors occur[5]. High-intensity flood attacks like Mirai, HTTP Flood and DDoS TCP remain clearly separable from one another, with only minimal cross-confusion between them. The primary confusion patterns occur between behaviourally quieter attacks, SlowLoris and Reconnaissance OS Scan

[5]Corresponding confusion matrices and per-class performance tables for the Nest Indoor and Netatmo camera are provided in Appendix F

Table 4.7: Multi-Class Anomaly Detection Results for Arlo Q Indoor Camera (60-Second Window).

Class	Precision	Recall	F1-score
Benign	0.9725	0.9381	0.9550
Mirai	0.9386	0.9469	0.9427
HTTP Flood	0.9725	0.9381	0.9550
DDoS TCP	1.0000	0.9649	0.9821
SlowLoris	0.9381	0.9381	0.9381
Recon OS Scan	0.9369	0.9204	0.9286
DNS Spoofing	0.8819	0.9825	0.9295
Accuracy			**0.9470**
Macro Avg	0.9486	0.9470	0.9473
Weighted Avg	0.9486	0.9470	0.9473

share 11 misclassifications between them, and DNS Spoofing shows small amounts of error from multiple other classes. These patterns suggest that the compact nine-feature representation captures sufficient information to distinguish between high-intensity disruption attacks with distinct signatures, but that attacks designed to produce subtle or low-volume traffic patterns retain some behavioural overlap.

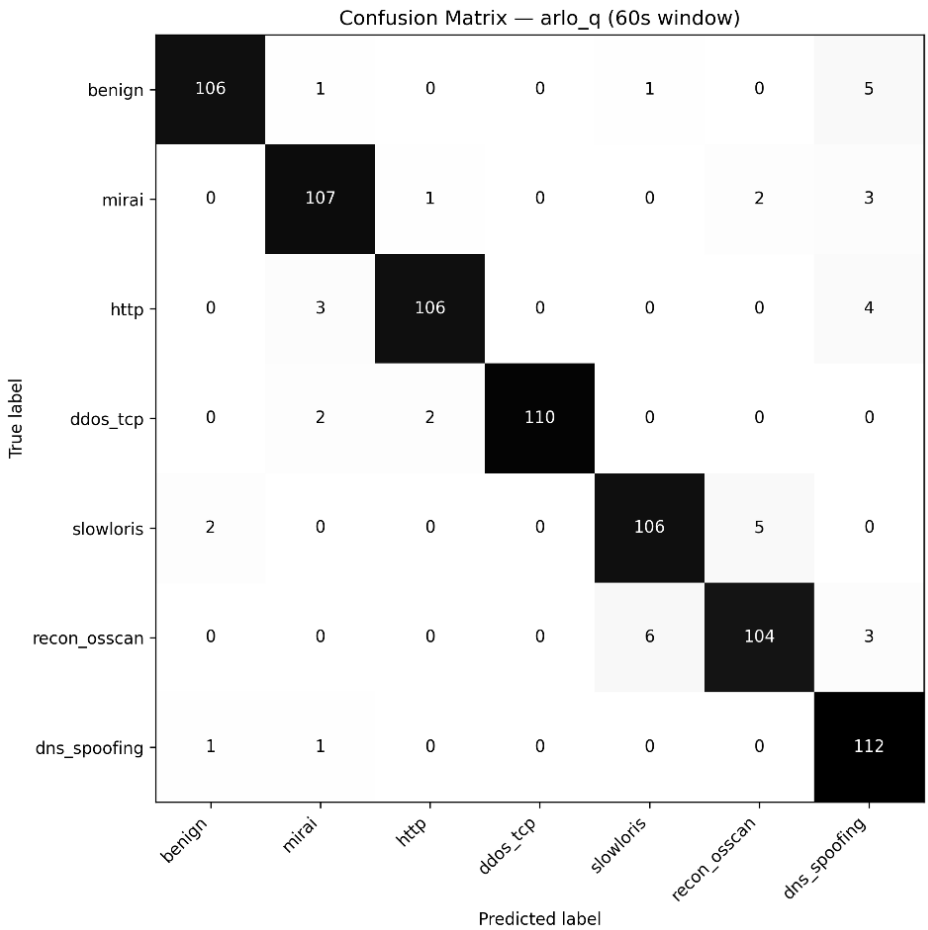

Figure 4.1: Confusion matrix for ARLO Q Indoor Camera.

4.5 Discussion

The Device Identification and Anomaly Detection (DIAD) framework is the primary artifact in this study. Proposed in Chapter 3 (Section 3.6) and implemented through the staged evaluation presented in this chapter, its tested form is shown in Figure 4.2. This section brings the results together to assess the framework itself, describing what the evaluation demonstrates about its viability and how the tested version differs from the proposed version that entered this chapter. The results support the framework's core design while also exposing conditions under which its performance varies. Sections 4.5.1 through 4.5.3 work through the framework stage by stage, and Section 4.5.4 covers further observations that span multiple stages.

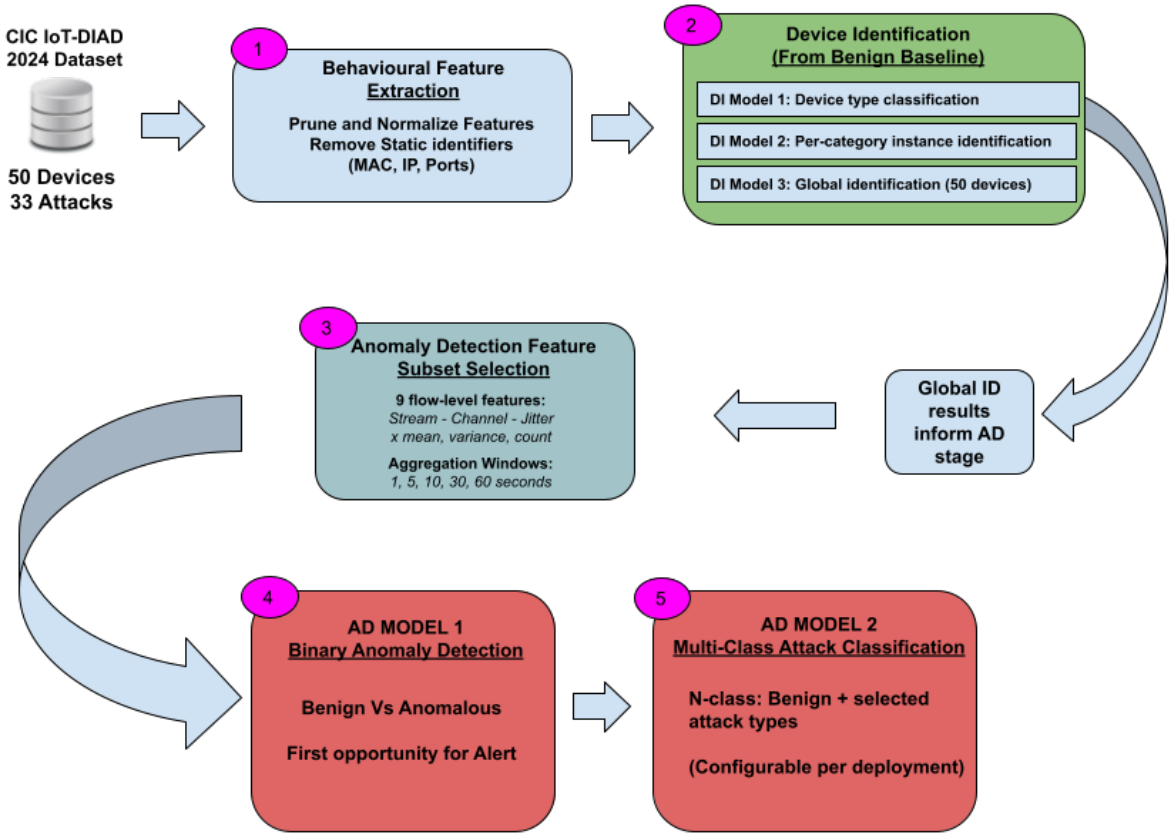

Figure 4.2: The evaluated DIAD framework, presented as a configurable pipeline. The stage structure is dataset-agnostic in principle; devices, attacks, and aggregation windows can be specified per deployment."Stages" are noted numerically in the top-left of their respective boxes.

The structural stages of the framework remained unchanged between the proposed and tested versions: feature extraction, device ID, AD feature subset selection, binary AD, and multi-class attack classification. What changed is framing. The proposed version (Chapter 3, Figure 3.1) described the specific experimental configuration used in this study. Three cameras, six attacks, nine AD features, and five temporal aggregation windows. Figure 4.2 presents the framework's evaluated form in a configurable state, with the stage structure

introduced but study-specific descriptions generalized. These differences highlight the goal of creating a more elastic framework that may benefit from the expansion of features and attack data.

4.5.1 Stages 1-2: Feature Extraction and Device Identification

Across all three DI configurations, performance degraded gradually as classification granularity increased. The drop wasn't sharp; it simply reflected device similarity. The more like-devices behaved similarly on the network, the harder they were to tell apart. Obvious, perhaps, but worth noting. Device-type classification indicated that device categories produce distinct communication patterns, with instance-level identification within categories revealing an expected behavioural overlap. This overlap is seen most clearly in low-throughput plug devices. Misclassifications primarily occurred where this behaviour was similar, for example, in identical smart plug families, not randomly across the label space. This could mean that the models succeeded in learning the behavioural structure and not the non-flow-related artifacts of the dataset. The strict removal of static identifiers (MAC, IP, vendor strings) reinforces these points and links back with prior findings that IoT devices must be identified through communication behaviour and avoid spoofable attributes (Okey et al., 2025; Xu et al., 2025).

The comparison between DI Model 1, Model 2, and Model 3 validates the first stage of the proposed DIAD process and part of the primary objective of this research: *device-specific behavioural baselines can be established in a scalable way using a reduced set of flow-based features*.

4.5.2 Stage 3: AD Feature Subset Selection

All traffic passed into the anomaly detection stage - both the benign baseline traffic and the attack traffic it is compared against - was represented using the same compact set of nine flow-level descriptors, derived from stream, channel, jitter x mean, variance and count. These 9 descriptors are then summarized across five temporal aggregation windows. This shared representation was then used as input for both anomaly detection models that followed, supplying Binary AD (Stage 4) and Multi-Class AD (Stage 5) from a single preparation step.

Despite these reductions, AD performance remained stable across both models. This supports prior research showing that carefully selected behavioural features can preserve detection performance while reducing computational overhead (Okey et al., 2025). The experiments here back up that feature quality and representational structure matter more than raw feature quantity. Increasing feature table size alone does not guarantee improved separability, particularly when behaviours overlap (Kumar et al., 2025). From an implementation perspective, this is very relevant because detection systems deployed in complex IoT environments must balance accuracy with efficiency. Lightweight models using compacted flow

statistics are more realistic than feature-heavy or payload-dependent approaches (Ponnumani et al., 2026).

4.5.3 Stages 4-5: Binary and Multi-Class Anomaly Detection

Binary anomaly detection showed that modelling device-specific benign behaviour allows for a reliable foundation for detecting anomalous deviations in data flow. Performance remained consistently high across all aggregation windows, with most combinations reaching 99% accuracy or above at moderate aggregation windows. The compact nine-feature selection for this binary decision task proved effective for this task, reinforcing the argument that minimal, well-chosen features can support reliable detection in resource-limited environments.

However, future analysis of feature selection may further assist performance. Analytically evaluating the addition or removal of certain descriptors could help identify a minimal yet optimal feature subset, improve efficiency while maintaining or potentially enhancing detection accuracy.

Multi-class attack classification presented a more complex scenario. Performance remained strong on devices with robust baseline traffic (Nest Indoor (96.26%) and Arlo Q (94.70%) — but dropped noticeably on Netatmo (85.93%). Benign traffic remained clearly separable from anomalous activity, but confusion between behaviourally similar attack types was also clearly present in all three cases. This indicates that high-intensity disruption attacks can produce overlapping flow dynamics when represented through a limited flow-based and compact feature set. This alone does not write off the method. Instead, it shows a distinction between detection and differentiation: detecting anomalous traffic is more straightforward than determining the precise type of that same anomalous traffic.

These differentiation difficulties align with prior published research that even in high-performing detection frameworks encounter challenging categories and more complex attack types where feature distributions exhibit potential overlap (Bensaoud & Kalita, 2025). In this context, the confusion observed between the attack types does not indicate model instability or insufficiency but instead reflects overlapping statistical characteristics under a limited flow-based representation. These confusions were largely attack categories confused amongst themselves, and not significantly confusing benign traffic with anomalous. This solidifies the model's viability as a first-stage anomaly detection mechanism, even if finer-grained attack identification may require expanded or more specialized feature representations.

4.5.4 Further Observations

The Role of Baseline Volume

A consistent pattern across both binary and multi-class anomaly detection results was the influence of benign baseline volume on classification performance. By chance, the three cam-

eras evaluated represent a natural increase for comparison in baseline availability with Netatmo (142 samples per class), Arlo Q (566 samples per class) and Nest Camera (993 samples per class). The resulting multi-class accuracies of 85.93%, 94.70%, and 96.26% show a roughly 10-point gap between the best and worst performing devices, with a clear relationship between baseline volume and performance. This has implications for any real-world deployment. IoT devices with low-traffic volume, such as sensors or devices with long sleep cycles, may require extended baseline capture periods before anomaly detection can be applied with confidence. The Netatmo confusion matrix found in Appendix F illustrates this. Error patterns on the low-baseline device are broader, with benign traffic occasionally misclassified as DNS Spoofing. These are errors that do not appear on the higher-baseline devices, where confusion is contained to attack-versus-attack instead of attack-versus-benign. This distinction is important for security-minded deployment, as failing to detect an anomaly or attack carries a different risk than misclassifying between two attack types.

Window Size and Task Complexity

The aggregation window analysis revealed different optimal behaviours for binary and multi-class detection. Binary detection showed attack-dependent optimal windows, with some attacks reaching strong performance at shorter windows and others requiring longer observation periods to gather sufficient behavioural signatures. Multi-class classification, though, showed an across-the-board preference for longer windows, with 60-second aggregation producing the best results in every case. Distinguishing between seven classes requires a richer temporal context, and the compact feature representation benefits from longer observation periods when the task is more demanding. Again, from a deployment perspective, this says the aggregation windows could be adapted based on the detection stage. For example, shorter windows for rapid binary flagging, and longer windows for detailed attack classification.

4.6 Chapter Summary

The results in this chapter align directly with the primary objective of developing and evaluating a conceptual framework for IoT device identification and anomaly detection using behavioural traffic modelling. The DI models showed that the selected packet and flow-level behavioural features can distinguish devices at multiple levels, with 95% + accuracy, ranging from broad device type categories down to 50 individual devices. The anomaly detection models pushed this approach across three camera devices and six attack types, with binary detection exceeding 98% accuracy and multi-class classification reaching up to 96.26% accuracy on the device with the most baseline data, all while limited to only 9 flow-level features.

Through this process, the secondary objectives were addressed. The characteristics of benign and malicious traffic within the CIC IoT-DIAD 2024 dataset were analyzed in detail, with

device behavioural profiles established under benign conditions before attack traffic was introduced. Existing approaches to IoT device profiling and anomaly detection were examined through the literature review and informed the design of the framework. The effectiveness of a detection pipeline was then evaluated using the dataset as a benchmark, with results demonstrating strong performance across both device identification and anomaly detection tasks. Overall, the findings indicate that a behaviour-driven approach to IoT device identification and anomaly detection can provide a reproducible foundation for further refinement and study.

5 Conclusion

5.1 Introduction

IoT security is a problem defined by scale and diversity, and the challenge of detecting threats across heterogeneous device networks is not one that any single existing approach handles well. This study set out to evaluate whether a hybrid behavioural framework, combining packet-level device identification with flow-level anomaly detection, could offer a practical and scalable response to that challenge. The preceding chapters built the case for that approach, examined the existing literature, described the methodology, and presented the results. This chapter brings those threads together, starting with a summary of the research.

5.2 Summary of Research

The first section of this report established that IoT devices represent a significant security risk. As integral components of smart home environments and production systems, their exploitation can carry serious economic and physical consequences. While combined existing security approaches provide a well-rounded baseline, the limitations of current detection methods highlight the need for a more dynamic and scalable framework capable of flagging anomalies at an earlier stage.

The literature review covered the foundational work of Rabbani et al. (2025) and the Canadian Institute of Cybersecurity. Their development of the CIC IoT-DIAD 2024 dataset provided the benchmark for this entire study, and their hybrid packet and flow-based framework

on the previous year's dataset established that combining both traffic layers can outperform either method alone. The review also revealed that purposefully selected behavioural features and avoiding static identifiers provided the most solid base for both device identification and anomaly detection in heterogeneous IoT environments. Among the machine learning classifiers evaluated across the literature, Random Forest emerged as a consistently strong performer for both tasks, offering resilience to class imbalance and reliable performance at scale across diverse IoT device networks.

The methodology followed the hybrid DIAD pipeline skeleton provided by the literature, implementing Random Forest classifiers at each stage. While the framework was designed to be as lightweight as possible, practical challenges around model size and available processing power required the development of a memory-friendly training configuration. These included per-device sample caps and reduced tree complexity for the global device identification model. Feature importance analysis proved particularly valuable throughout this process. It revealed which behavioural features carried the most weight and shaped the decision to reduce the anomaly detection input to just 9 flow descriptors derived from stream, channel and jitter statistics.

The results showed a strong performance across both stages of the pipeline. Device identification achieved an overall 95.3% accuracy across all devices simultaneously. Anomaly detection was evaluated across three camera devices: The Arlo Q Indoor Camera, the Nest Indoor Camera, and the Netatmo Camera, against six attack types selected to provide coverage across five CIC IoT-DIAD 2024 attack categories. Binary detection reached 99%+ accuracy across most device -attack combinations at moderate aggregation windows, and multi-class classification reached 96.26% accuracy on the Nest Indoor Camera at the 60-second window, using only 9 flow-level features. Performance varied with benign baseline volume across devices, with the Netatmo Camera producing weaker results than the other two. This was a finding that reinforces the importance of sufficient baseline data availability in behavioural detection. The overall findings validated the design decisions made throughout the project and indicated clear paths for further refinement.

5.3 Research Objectives

The secondary objectives of this study were designed to build progressively toward the primary objective. Each one addressed a component of the broader research question, and together they proved the foundation from which the primary objective could be evaluated and met.

5.3.1 Secondary Objective 1

"Investigate the security challenges and vulnerabilities represented in the CIC IoT-DIAD 2024 dataset."

The CIC IoT-DIAD 2024 dataset provided a comprehensive representation of the security challenges facing modern IoT environments. IoT devices are particularly vulnerable compared to traditional networked systems. Limited processing capacity, memory constraints, and weak built-in security mechanisms mean that many devices lack the ability to support host-based defences. The dataset captured this reality across 50 unique devices, exposing them to 33 distinct attacks across 7 categories. Signature-based systems such as Snort remain valuable as part of a layered defence strategy, providing reliable detection of *known* external threats at the network perimeter. However, their reliance on predefined rule sets means that they are poorly suited to detecting *behavioural* anomalies that may only become visible through deeper traffic analysis, especially in the context of novel or zero-day attacks. This objective was addressed in Sections 1.2 and 2.2.

Understanding the nature and scale of these vulnerabilities led to examining what existing detection approaches were doing about them, and where the gaps remained.

5.3.2 Secondary Objective 2

"Examine existing approaches to IoT device profiling and anomaly detection as applied to real network traffic."

The literature review examined both established and emerging approaches to IoT device profiling and anomaly detection. Signature-based intrusion detection systems have been widely deployed for decades and continue to provide value, but their limitations in heterogeneous IoT environments are well known. Flow-based anomaly detection approaches have demonstrated strong independent results, and packet-based device fingerprinting has shown reliable performance across diverse device populations. At the same time, no standalone hybrid framework combining both layers had been developed prior to the work of Rabbani et al. (2025), whose integrated packet and flow-based pipeline established that combining both approaches significantly outperforms either method alone. This finding directly informed the design of the DIAD pipeline evaluated in this study. This objective was addressed in Sections 2.2, 2.3 and 2.4.

With the foundation established, the next step was to apply that understanding directly to the dataset and analyze what the traffic data actually revealed about device behaviour under both benign and attack conditions.

5.3.3 Secondary Objective 3

"Analyze the characteristics of benign and malicious traffic within the dataset to distinguish device behaviours across multiple device types and attack categories."

The characteristics of both benign and malicious (anomalous) traffic were analyzed throughout the implementation of the DIAD pipeline. For device identification, feature importance

analysis revealed the most discriminative packet-level features, while leakage-prone static identifiers were deliberately removed. For anomaly detection, the success of stream, channel and jitter metrics in prior research informed the decision to focus the AD feature set on those three indicators alone, combined with their statistical mean, variance and count across temporal aggregation windows. Despite this intentional reduction to just 9 features, binary and multi-class detection results remained strong. The analysis also suggested that further refinement through progressive feature addition or attack-specific feature selection could improve differentiation between behaviourally similar attack types. This objective was addressed in Sections 3.6, 4.3 and 4.4.

This analysis directly enabled the fourth objective, providing the feature understanding and baseline behavioural profiles needed to evaluate the framework against labelled attack traffic.

5.3.4 Secondary Objective 4

"Evaluate the effectiveness of a scalable detection approach using the CIC IoT-DIAD 2024 dataset as a benchmark."

The CIC IoT-DIAD 2024 dataset served as the primary testbed for evaluation throughout this study, with the results of the CIC (Rabbani et al., 2025) on the previous year's dataset providing the broader performance benchmarks against which findings could be compared and contextualized.

Device identification was evaluated across three progressive models. At the category level, the classifier achieved 99.7% accuracy, confirming that broad device families produce distinct traffic patterns. Instance-level identification within categories reached between 87.6% and 99.9% depending on device family, with the most significant drop occurring among low-throughput plug devices that share highly similar behavioural signatures. This result indicated a genuine overlap in device behaviour and perhaps not model instability. Global identification across all devices simultaneously achieved 95.3% accuracy, with performance degrading gradually and predictably as classification granularity increased. This pattern supports the scalability of the behavioural approach.

Anomaly detection across three camera devices demonstrated reliable binary detection, with most device-attack combinations exceeding 99% accuracy on the Nest Indoor Camera, 94.70% on the Arlo Q, and 85.93% on the Netatmo at the 60-second window. Multi-class classification at the 60-second window was strongest on devices with richer benign baselines, raching 96.26% on the Nest Indoor Camera and 94.70% accuracy on the Arlo Q, with the lower baseline Netatmo Camera reaching 85.93%, all using only 9 flow-level features.The clear relationship between benign baseline volume and classification performance identifies baseline quality (or quantity) as a reliable predictor of detection effectiveness. These results confirm that a behaviour-driven pipeline using intentionally selected flow-level features can

serve as an effective and scalable detection framework, while also indicating practical deployment considerations around baseline capture. This objective was addressed in Sections 4.3 and 4.4.

With each secondary objective addressed, the study was in a position to evaluate whether the primary objective had been achieved.

5.3.5 Primary Objective

"Develop and evaluate a conceptual framework for scalable IoT device identification and anomaly detection based on behavioural traffic modelling."

The primary objective of this study was met. A two-stage Device Identification and Anomaly Detection (DIAD) framework was developed and evaluated using the CIC IoT-DIAD 2024 dataset. While the individual components of the pipeline were implemented and evaluated as separate stages and not as a single fully integrated system, each stage was tested independently and produced strong, consistent results. Device ID was evaluated across 3 progressive classification models, resulting in global identification of 50 devices simultaneously at 95.3% accuracy.

Anomaly detection was evaluated across three camera devices against six attack types selected to cover five CIC IoT-DIAD 2024 attack categories. Both binary and multi-class classification were evaluated, with binary detection reaching 99%+ accuracy across most device-attack combinations and multi-class classification reaching between 85.93% and 96.26% accuracy depending on device baseline volume, all using only 9 flow-level features. While the anomaly detection evaluation was deliberately limited in scope, the results from this focused case study were sufficiently promising to support the broader framework design. The findings demonstrate that the main components of a scalable behaviour-driven IoT security framework are technically viable and can be practically implemented with a lean and efficient feature representation. This objective was addressed throughout Chapters 3 and 4.

5.4 Reflection

The project's direction shifted noticeably between planning and completion. The original scope centered on a single device versus three attack types as a proof of concept for anomaly detection, expecting that extending the evaluation beyond one device would exceed the time and hardware available. In practice, the opposite proved true. The device identification work scaled more easily than anticipated, and the time saved there created room to expand anomaly detection across three cameras and six attack types. The stronger findings around baseline volume, one of the useful contributions, only became visible because of that expansion and would have been missed under the original scope.

Much of the value of the work sat outside the modelling itself. Dataset preparation, class balancing, feature selection and reproducibility management took far more time than model training, and were the areas where most of the practical lessons were learned. Running models across multiple machines with questionable archiving practices produced early problems that had to be corrected before results could be trusted. These are details that don't really appear in published work but shape the outcome substantially.

Looking back, the decision to intentionally reduce the anomaly detection feature set to nine descriptors instead of replicating the full Rabbani et al. (2025) pipeline carried some risk. It could have underperformed and undermined the case for the hybrid approach. Thankfully, it did not, and the performance remained strong across multiple devices and attack types, and ultimately became the most satisfying outcome of the project.

5.5 Research Contribution

This study contributed to the work on hybrid behavioural approaches to IoT security. We added validation and reinforced the findings of Rabbani et al. Rabbani et al. (2025), showing that a combined packet and flow-based framework for device ID and AD produces strong results when applied to the CIC IoT-DIAD 2024 dataset.

Aside from the replication of the work done by the team at the CIC, the study introduced a minimal anomaly detection feature set, reducing the input to just 9 flow descriptors derived from stream, channel and jitter statistics. The strong performance maintained under this reduced set highlights the argument that feature quality matters more than feature quantity in behavioural IoT anomaly detection. This has practical implications for real-world IoT network deployment, where lightweight models are more useful in resource-constrained environments.

Additionally, the temporal window analysis showed different optimal behaviours for binary and multi-class detection. For binary detection, the optional aggregation duration was attack-specific, while multi-class classification showed a uniform preference for longer windows across all three devices. This finding could assist with the design of adaptive detection systems that vary the window size based on the detection stage.

5.6 Future Work

A natural next step would be expanding the anomaly detection evaluation beyond the camera category. Cameras were selected for this study due to their position as the highest-traffic device category in the dataset, which allowed meaningful evaluation across multiple devices and attack types. However, extending the AD pipeline eventually across all 50 devices in the dataset would provide a stronger basis for claims about scalability. This expansion would need to account for the fact that not all attack categories affected all devices equally in the

dataset, meaning that device selection for AD evaluation would need to note which attack types generated sufficient traffic volume against each specific device to support meaningful classification with the models developed here. Additionally, an expansion of attack coverage would be valuable in future work. The six attack types selected for this study were chosen for their high-throughput communication patterns, which produced the most data for evaluation. Testing the same pipeline against the remaining attack categories in the dataset would establish whether the compact feature set remains effective across a broader threat landscape.

Future work may also include the combination of the models developed in this study, using more capable hardware or cloud-based computing to remove the low-memory constraints and allow for a more comprehensive evaluation without the memory-saving trade-offs applied in the global device identification model.

Beyond scope expansion, there is also room for optimizing the models themselves. This study used Random Forest with default parameters (aside from a fixed 300 trees and standard imputation), prioritizing a clean baseline over tuned performance. Hyperparameter tuning, further fine-grained per-device feature importance analysis, and alternative classifier comparisons (XGBoost or lightweight neural networks) could also refine the multi-class results

Finally, the ultimate extension of this work would be expanding and testing the validated pipeline against live network traffic in a real or simulated heterogeneous IoT environment. This could include direct integration with a live feature extraction pipeline to feed traffic into the classification models in real time, removing the dependency on pre-processed CSV files and moving the framework closer to a deployable detection system.

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A

CIC IoT-DIAD 2024 Device List

Table A.1: List of CICIoT-DIAD 2024 Devices by Category (Grouped Layout)

Camera Devices	Audio Devices	Home Automation
AMCREST WiFi Camera	Amazon Echo Dot 1	Atomi Coffee Maker
Arlo Base Station	Amazon Echo Dot 2	Cocoon Smart HVAC Fan
Arlo Q Indoor Camera	Amazon Echo Show	Govee Smart Humidifier
DCS8000LHA1 D-Link Mini Camera	Amazon Echo Studio	LG Smart TV
HeimVision Smart WiFi Camera	Google Nest Mini Speaker	Levoit Air Purifier
HeimVision SmartLife Radio/Lamp	Sonos One Speaker	iRobot Roomba
Home Eye Camera	harman kardon (Ampak Technology)	**Lighting**
Nest Indoor Camera	**Power Outlets**	LIFX Lightbulb
Netatmo Camera	Amazon Plug	LampUX RGB
Rbcior Camera	GoSund Bulb	Lumiman Bulb
TP-Link Tapo Camera	GoSund Power Strip (1)	Teckin Light Strip
Yi Indoor 2 Camera	GoSund Power Strip (2)	**Hub Devices**
Yi Indoor Camera	GoSund Smart Plug WP2 (1)	AeoTec Smart Home Hub
Yi Outdoor Camera	GoSund Smart Plug WP2 (2)	Eufy HomeBase 2
	GoSund Smart Plug WP2 (3)	Philips Hue Bridge
	GoSund Smart Plug WP3 (1)	SmartThings Hub
	GoSund Smart Plug WP3 (2)	**Sensor**
	Teckin Plug 1	Netatmo Weather Station
	Teckin Plug 2	
	Wemo Smart Plug (Wemo.Mini.4A3)	
	Yutron Plug 1	
	Yutron Plug 2	

B

DI Model 2 Per-Category Results

This appendix contains the full per-device classificaiton results for DI Model 2, organized by device category. Each table reports precision, recall, F1-score, and support for every device within that category, as referenced in Section 4.3.2.

B.1 Audio Device

Table B.1: Per-device classification performance for Audio Device category (Random Forest, 80/20 Train/Test).

Device	Precision	Recall	F1-score	Support
Amazon Echo Dot 1	0.9838	0.9932	0.9885	1,472
Amazon Echo Dot 2	0.9982	0.9987	0.9985	6,248
Amazon Echo Show	0.9961	0.9944	0.9953	8,642
Amazon Echo Studio	0.9862	0.9873	0.9867	2,602
Google Nest Mini Speaker	0.9996	0.9985	0.9991	2,722
Sonos One Speaker	0.9916	0.9874	0.9895	478
harman kardon (Ampak Technology)	0.9979	0.9974	0.9976	1,901
Accuracy			**0.9953**	**24,065**
Macro Avg	0.9934	0.9939	0.9936	24,065
Weighted Avg	0.9953	0.9953	0.9953	24,065

B.2 Camera

Table B.2: Per-device classification performance for Camera category (Random Forest, 80/20 Train/Test).

Device	Precision	Recall	F1-score	Support
AMCREST WiFi Camera	0.9921	0.9984	0.9952	627
Arlo Base Station	0.9950	0.9917	0.9934	603
Arlo Q Indoor Camera	0.9999	1.0000	0.9999	49,132
D-Link Mini Camera	1.0000	1.0000	1.0000	101
HeimVision Smart WiFi Camera	1.0000	0.9985	0.9992	1,322
HeimVision SmartLife	1.0000	1.0000	1.0000	755
Home Eye Camera	1.0000	1.0000	1.0000	2,087
Nest Indoor Camera	1.0000	1.0000	1.0000	14,467
Netatmo Camera	0.9965	0.9993	0.9979	1,427
Rbcior Camera	1.0000	0.9991	0.9996	1,152
TP-Link Tapo Camera	0.9977	0.9953	0.9965	858
Yi Indoor 2 Camera	0.9989	0.9772	0.9879	1,882
Yi Indoor Camera	0.9159	0.9567	0.9359	854
Yi Outdoor Camera	0.8927	0.8860	0.8893	535
Accuracy			**0.9979**	**75,802**
Macro Avg	0.9849	0.9859	0.9853	75,802
Weighted Avg	0.9980	0.9979	0.9980	75,802

B.3 Home Automation

Table B.3: Per-device classification performance for Home Automation category (Random Forest, 80/20 Train/Test).

Device	Precision	Recall	F1-score	Support
Atomi Coffee Maker	1.0000	1.0000	1.0000	658
Cocoon Smart HVAC Fan	1.0000	0.9982	0.9991	557
Govee Smart Humidifer	0.9899	1.0000	0.9949	98
LG Smart TV	0.9990	1.0000	0.9995	1,990
Levoit Air Purifier	1.0000	0.9978	0.9989	449
iRobot Roomba	1.0000	0.9924	0.9962	131
Accuracy			**0.9992**	**3,883**
Macro Avg	0.9981	0.9981	0.9981	3,883
Weighted Avg	0.9992	0.9992	0.9992	3,883

B.4 Hub Device

Table B.4: Per-device classification performance for Hub Device category (Random Forest, 80/20 Train/Test).

Device	Precision	Recall	F1-score	Support
AeoTec Smart Home Hub	0.8477	0.8843	0.8656	1,366
Eufy HomeBase 2	1.0000	1.0000	1.0000	961
Philips Hue Bridge	1.0000	1.0000	1.0000	2,479
SmartThings Hub	0.8851	0.8487	0.8665	1,434
Accuracy			**0.9399**	**6,240**
Macro Avg	0.9332	0.9333	0.9330	6,240
Weighted Avg	0.9403	0.9399	0.9399	6,240

B.5 Lighting

Table B.5: Per-device classification performance for Lighting category (Random Forest, 80/20 Train/Test).

Device	Precision	Recall	F1-score	Support
LIFX Lightbulb	1.0000	1.0000	1.0000	153
LampUX RGB	1.0000	1.0000	1.0000	909
Lumiman bulb	0.9956	0.9985	0.9970	674
Teckin Light Strip	0.9985	0.9954	0.9970	657
Accuracy			**0.9983**	**2,393**
Macro Avg	0.9985	0.9985	0.9985	2,393
Weighted Avg	0.9983	0.9983	0.9983	2,393

B.6 Power Outlets

Table B.6: Per-device classification performance for Power Outlets category (Random Forest, 80/20 Train/Test).

Device	Precision	Recall	F1-score	Support
Amazon Plug	1.0000	1.0000	1.0000	146
GoSund Bulb	0.8091	0.8768	0.8416	909
GoSund Power Strip (2)	0.8645	0.8141	0.8385	909
GoSund Smart Plug WP2 (2)	0.8502	0.8240	0.8369	909
GoSund Smart Plug WP3 (1)	0.8011	0.8319	0.8162	910
GoSund Smart Plug WP2 (1)	0.8434	0.9538	0.8952	909
GoSund Smart Plug WP2 (3)	0.8548	0.7899	0.8210	909
GoSund Power Strip (1)	0.8335	0.8637	0.8484	910
GoSund Smart Plug WP3 (2)	0.8920	0.7811	0.8328	909
Teckin Plug 1	0.9620	0.9732	0.9676	598

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Device	Precision	Recall	F1-score	Support
Teckin Plug 2	0.9476	0.9554	0.9515	606
Wemo Smart Plug (Wemo.Mini.4A3)	1.0000	1.0000	1.0000	153
Yutron Plug 1	0.9773	0.9678	0.9725	622
Yutron Plug 2	0.9520	0.9429	0.9475	631
Accuracy			**0.8755**	**10,030**
Macro Avg	0.8991	0.8982	0.8978	10,030
Weighted Avg	0.8767	0.8755	0.8751	10,030

C

DI Model 3 Full Results

This appendix contains the per-device performance for the global 50-device classification model discussed in Section 4.3.3.

Table C.1: Per-device classification performance for IoT device identification (Random Forest, 100k samples maximum per device, 80/20 Train/Test).

Device	Precision	Recall	F1-score	Support
AMCREST WiFi Camera	0.9743	0.9681	0.9712	627
AeoTec Smart Home Hub	0.7393	0.8572	0.7939	1366
Amazon Echo Dot 1	0.9695	0.9056	0.9364	1472
Amazon Echo Dot 2	0.9981	0.9984	0.9982	6248
Amazon Echo Show	0.9949	0.9977	0.9963	8642
Amazon Echo Studio	0.9614	0.9854	0.9732	2602
Amazon Plug	0.9779	0.9110	0.9433	146
Arlo Base Station	0.9764	0.9585	0.9674	603
Arlo Q Indoor Camera	0.9997	0.9997	0.9997	20000
Atomi Coffee Maker	0.9139	0.9362	0.9249	658
Cocoon Smart HVAC Fan	0.9964	0.9892	0.9928	557
D-Link Mini Camera	1.0000	1.0000	1.0000	101
Eufy HomeBase 2	0.9802	0.9792	0.9797	961
GoSund Bulb	0.6735	0.6172	0.6441	909
GoSund Power Strip (2)	0.7048	0.7668	0.7345	909

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Device	Precision	Recall	F1-score	Support
GoSund Smart Plug WP2 (2)	0.8837	0.6018	0.7160	909
GoSund Smart Plug WP3 (1)	0.6977	0.7505	0.7231	910
GoSund Smart Plug WP2 (1)	0.5732	0.8229	0.6757	909
GoSund Smart Plug WP2 (3)	0.7907	0.6942	0.7393	909
Google Nest Mini Speaker	0.9894	0.9901	0.9897	2722
GoSund Power Strip (1)	0.5997	0.7580	0.6696	909
GoSund Smart Plug WP3 (2)	0.8353	0.6865	0.7536	909
Govee Smart Humidifier	0.9892	0.9388	0.9634	98
HeimVision Smart WiFi Camera	1.0000	0.9985	0.9992	1322
HeimVision Radio/Lamp	0.9625	0.9868	0.9745	755
Home Eye Camera	0.9986	0.9986	0.9986	2087
LG Smart TV	0.9321	0.9799	0.9554	1990
LIFX Lightbulb	0.9801	0.9673	0.9737	153
LampUX RGB Light	0.9025	0.7129	0.7966	909
Levoit Air Purifier	0.9933	0.9978	0.9956	449
Lumiman Bulb	0.9338	0.9837	0.9581	674
Nest Indoor Camera	1.0000	1.0000	1.0000	14467
Netatmo Camera	0.9832	0.9860	0.9846	1427
Netatmo Weather Station	0.9776	0.9987	0.9880	786
Philips Hue Bridge	0.9947	0.9939	0.9943	2478
Rbcior Camera	0.9587	0.9670	0.9628	1152
SmartThings Hub	0.8363	0.7517	0.7918	1434
Sonos One Speaker	0.9675	0.8724	0.9175	478
TP-Link Tapo Camera	0.9666	0.9779	0.9722	858
Teckin Light Strip	0.9880	0.8767	0.9290	657
Teckin Plug 1	0.9727	0.8344	0.8983	598
Teckin Plug 2	0.8535	0.9323	0.8912	606
Wemo Smart Plug Mini (4A3)	0.9850	0.8562	0.9161	153
Yi Indoor 2 Camera	0.9994	0.9538	0.9761	1882
Yi Indoor Camera	0.8083	0.9133	0.8576	854
Yi Outdoor Camera	0.7510	0.7103	0.7301	535
Yutron Plug 1	0.9390	0.9405	0.9398	622
Yutron Plug 2	0.8879	0.9033	0.8955	631
Harman Kardon	0.9963	0.9784	0.9873	1901
iRobot Roomba	0.9846	0.9771	0.9808	131

D

AD Model 1 Binary Anomaly Detection Full Results

This appendix contains the complete binary anomaly detection results across all three camera devices, six attack types, and five aggregation windows, as referenced in Section 4.4.1.

Table D.1: Binary anomaly detection results across all devices and attacks (Random Forest, 9 flow-level features, 80/20 Train/Test).

Device	Attack	Window (s)	Accuracy	Macro F1	Support	Errors
Arlo Q	Mirai	1	0.9903	0.9903	1,130	11
		5	0.9929	0.9929	1,130	8
		10	0.9938	0.9938	1,130	7
		30	0.9947	0.9947	1,130	6
		60	0.9956	0.9956	1,130	5
Arlo Q	HTTP Flood	1	0.9868	0.9868	227	3
		5	0.9956	0.9956	227	1
		10	0.9912	0.9912	227	2
		30	0.9912	0.9912	227	2
		60	0.9912	0.9912	227	2
Arlo Q	DDoS TCP	1	0.9940	0.9940	335	2
		5	1.0000	1.0000	335	0
		10	0.9940	0.9940	335	2

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Device	Attack	Window (s)	Accuracy	Macro F1	Support	Errors
		30	1.0000	1.0000	335	0
		60	1.0000	1.0000	335	0
Arlo Q	SlowLoris	1	0.9192	0.9192	20,792	1,680
		5	0.9789	0.9789	20,792	438
		10	0.9913	0.9913	20,792	181
		30	0.9987	0.9987	20,792	26
		60	0.9988	0.9988	20,792	24
Arlo Q	OSScan	1	0.9377	0.9377	28,672	1,786
		5	0.9921	0.9921	28,672	226
		10	0.9969	0.9969	28,672	88
		30	0.9980	0.9980	28,672	58
		60	0.9985	0.9985	28,672	44
Arlo Q	DNS Spoofing	1	0.9186	0.9186	3,268	266
		5	0.9841	0.9841	3,268	52
		10	0.9875	0.9875	3,268	41
		30	0.9924	0.9923	3,268	25
		60	0.9914	0.9914	3,268	28
Nest Indoor Camera	Mirai	1	0.9714	0.9714	3,321	95
		5	0.9952	0.9952	3,321	16
		10	0.9988	0.9988	3,321	4
		30	1.0000	1.0000	3,321	0
		60	0.9997	0.9997	3,321	1
Nest Indoor Camera	HTTP Flood	1	0.9718	0.9718	567	16
		5	0.9965	0.9965	567	2
		10	0.9947	0.9947	567	3
		30	1.0000	1.0000	567	0
		60	1.0000	1.0000	567	0
Nest Indoor Camera	DDoS TCP	1	0.9397	0.9397	398	24
		5	0.9975	0.9975	398	1
		10	0.9975	0.9975	398	1
		30	0.9975	0.9975	398	1
		60	0.9975	0.9975	398	1
Nest Indoor Camera	SlowLoris	1	0.8843	0.8843	6,614	765
		5	0.9881	0.9881	6,614	79
		10	0.9915	0.9915	6,614	56
		30	0.9952	0.9952	6,614	32
		60	0.9976	0.9976	6,614	16
Nest Indoor Camera	OSScan	1	0.8056	0.8055	13,572	2,639
		5	0.9868	0.9868	13,572	179

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Device	Attack	Window (s)	Accuracy	Macro F1	Support	Errors
		10	0.9940	0.9940	13,572	81
		30	0.9970	0.9970	13,572	41
		60	0.9995	0.9995	13,572	7
Nest Indoor Camera	DNS Spoofing	1	0.9303	0.9303	7,734	539
		5	0.9948	0.9948	7,734	40
		10	0.9992	0.9992	7,734	6
		30	0.9997	0.9997	7,734	2
		60	0.9999	0.9999	7,734	1
Netatmo Camera	Mirai	1	0.9766	0.9766	3,028	71
		5	0.9851	0.9851	3,028	45
		10	0.9884	0.9884	3,028	35
		30	0.9888	0.9888	3,028	34
		60	0.9898	0.9898	3,028	31
Netatmo Camera	HTTP Flood	1	0.9385	0.9385	488	30
		5	0.9508	0.9508	488	24
		10	0.9549	0.9549	488	22
		30	0.9734	0.9734	488	13
		60	0.9795	0.9795	488	10
Netatmo Camera	DDoS TCP	1	0.9298	0.9298	57	4
		5	0.9123	0.9122	57	5
		10	0.9825	0.9824	57	1
		30	0.9825	0.9825	57	1
		60	1.0000	1.0000	57	0
Netatmo Camera	SlowLoris	1	0.9663	0.9663	1,958	66
		5	0.9729	0.9729	1,958	53
		10	0.9775	0.9775	1,958	44
		30	0.9847	0.9847	1,958	30
		60	0.9908	0.9908	1,958	18
Netatmo Camera	OSScan	1	0.9232	0.9232	1,472	113
		5	0.9436	0.9436	1,472	83
		10	0.9450	0.9450	1,472	81
		30	0.9592	0.9592	1,472	60
		60	0.9776	0.9776	1,472	33
Netatmo Camera	DNS Spoofing	1	0.9570	0.9570	2,769	119
		5	0.9689	0.9689	2,769	86
		10	0.9704	0.9704	2,769	82
		30	0.9827	0.9827	2,769	48
		60	0.9888	0.9888	2,769	31

* Netatmo DDoS TCP results are statistically limited (57 test samples). Interpret with caution.

E

AD Model 2: Row Counts for Cameras vs All Attacks

This appendix contains available attack traffic volume per device across all attack types in the CIC IoT-DIAD 2024 dataset. These counts informed the attack selection process for AD Model 2 (Section 3.6.3).

Table E.1: Mirai attack traffic rows per camera device.

Attack	Arlo Q	Nest Indoor	Netatmo
Mirai-greip Flood	2,824	8,301	7,570

Table E.2: DoS attack traffic rows per camera device.

Attack	Arlo Q	Nest Indoor	Netatmo
DoS HTTP Flood	566	1,416	1,218
DoS SYN Flood	321	292	82
DoS UDP Flood	629	891	385
DoS TCP Flood	227	600	184

Table E.3: DDoS attack traffic rows per camera device.

Attack	Arlo Q	Nest Indoor	Netatmo
DDoS SlowLoris	51,978	16,535	4,895
DDoS TCP Flood	837	993	142
DDoS ACK Fragmentation	11,447	2,948	1,937
DDoS HTTP Flood	797	1,090	183
DDoS ICMP Fragmentation	19,649	10,867	6,579
DDoS SynonymousIP Flood	134	204	142
DDoS UDP Flood	1,025	614	1,085
DDoS UDP Fragmentation	18,457	4,763	2,479

Table E.4: Reconnaissance attack traffic rows per camera device.

Attack	Arlo Q	Nest Indoor	Netatmo
Recon OS Scan	71,680	33,928	3,680
Recon Host Discovery	6,773	6,789	2,981
Recon Ping Sweep	1,624	1,969	2,806
Recon Port Scan	2,665	10,716	4,380
Vulnerability Scan	9,977	11,408	2,833

Table E.5: Spoofing attack traffic rows per camera device.

Attack	Arlo Q	Nest Indoor	Netatmo
DNS Spoofing	8,169	19,334	6,921
ARP Spoofing	34,857	11,751	2,510

Table E.6: Web-Based attack traffic rows per camera device.

Attack	Arlo Q	Nest Indoor	Netatmo
Backdoor Malware	191	1,706	2,838
Browser Hijacking	196	2,166	1,873
Command Injection	254	2,247	4,406
SQL Injection	193	2,520	909
Uploading Attack	77	662	656
XSS	236	2,220	3,540

Table E.7: BruteForce attack traffic rows per camera device.

Attack	Arlo Q	Nest Indoor	Netatmo
Dictionary BruteForce	0	0	0

F

Additional Results for AD Model 2

This appendix contains the confusion matrices and per-class performance tables for the Nest Indoor and Netatmo cameras at the 60-second window, complementing the Arlo Q analysis in Section 4.4.2.

F.1 Nest Indoor Camera

Table F.1: Multi-Class Anomaly Detection Results for Nest Indoor Camera (60-Second Window).

Class	Precision	Recall	F1-score
Benign	0.9442	0.9394	0.9418
Mirai	0.9548	0.9596	0.9572
HTTP Flood	0.9800	0.9849	0.9825
DDoS TCP	0.9655	0.9849	0.9751
SlowLoris	0.9453	0.9548	0.9500
Recon OS Scan	0.9844	0.9497	0.9668
DNS Spoofing	0.9648	0.9648	0.9648
Accuracy			**0.9626**
Macro Avg	0.9627	0.9626	0.9626
Weighted Avg	0.9627	0.9626	0.9626

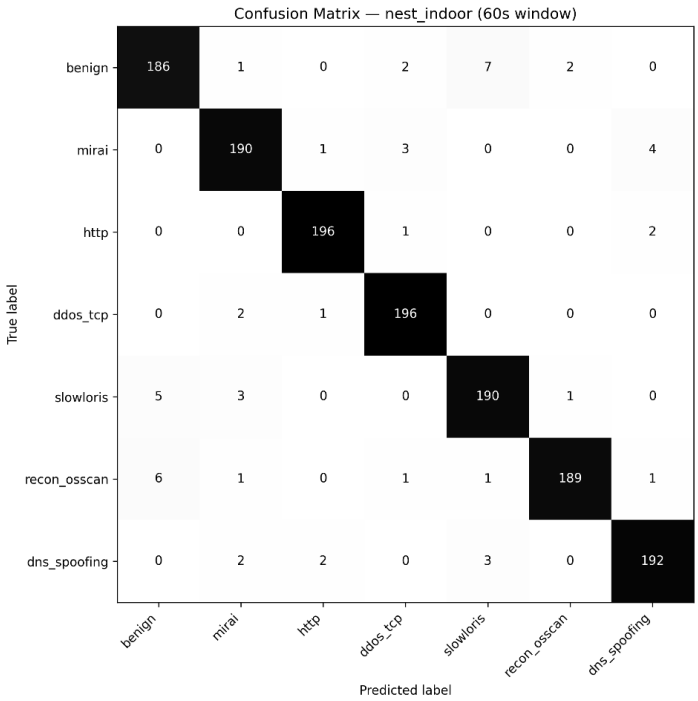

Figure F.1: Confusion matrix for Nest Indoor Camera (60-second window, 7-class multi-class classification). Overall accuracy: 96.26%, Macro F1: 96.26%.

F.2 Netatmo Camera

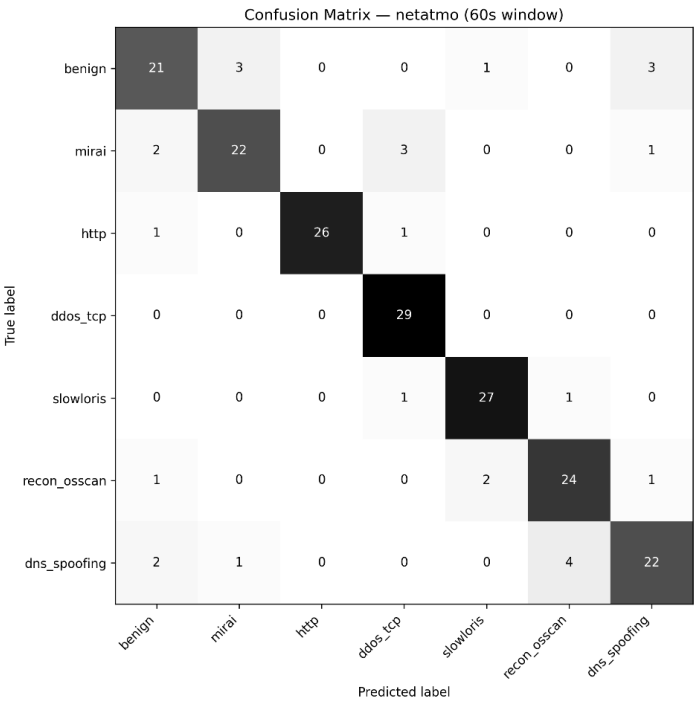

Figure F.2: Confusion matrix for Netatmo Camera (60-second window, 7-class multi-class classification). Overall accuracy: 85.93%, Macro F1: 85.79%.

Table F.2: Multi-Class Anomaly Detection Results for Netatmo Camera (60-Second Window).

Class	Precision	Recall	F1-score
Benign	0.7778	0.7500	0.7636
Mirai	0.8462	0.7857	0.8148
HTTP Flood	1.0000	0.9286	0.9630
DDoS TCP	0.8529	1.0000	0.9206
SlowLoris	0.9000	0.9310	0.9153
Recon OS Scan	0.8276	0.8571	0.8421
DNS Spoofing	0.8148	0.7586	0.7857
Accuracy			**0.8593**
Macro Avg	0.8599	0.8587	0.8579
Weighted Avg	0.8598	0.8593	0.8581

G

Machine Learning Models and Pipeline Artifact

This appendix provides direct access to the five machine learning models developed in this study and the high-resolution pipeline diagram that illustrates how they fit together. The five models comprise three Device Identification (DI) models and two Anomaly Detection (AD) models. All scripts use Random Forest classifiers and a fixed random seed of 42 for reproducibility. The DIAD pipeline artifact shows how these models combine into a single device identification and anomaly detection workflow.

Link to Git Hub Repository

https://github.com/coldui/ColDui01174_BPR_Scripts/tree/main

Word count metrics

NUC Bachelor Project Word Count:

Total Sum count: 16709 Words in text: 15957 Words in headers: 302 Words outside text (captions, etc.): 445 Number of headers: 85 Number of floats/tables/figures: 29 Number of math inlines: 1 Number of math displayed: 4
(errors:2) NOTE: References are excluded.